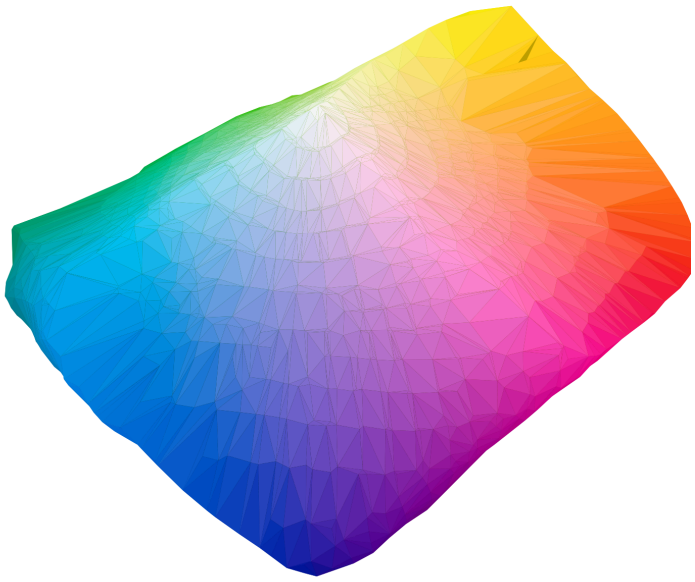
USER HANDBOOK

InkProf

User Handbook

Quality Profiling RGB printer

Profiling, measurement and verification of RGB printers



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USER HANDBOOK | v1.1.1 | 2026-10-13

What is InkProf?

InkProf is an open-source MATLAB app for creating, checking and improving ICC profiles for RGB-driven printers.

A profile for your printing combination

An ICC profile describes colour reproduction for a specific combination of printer, paper, ink and print settings. InkProf creates colour targets, collects spectral measurements and builds the profile using ArgyllCMS. You can also verify an existing ICC profile without building a new one.

A connected, traceable workflow

Targets, settings, measurements, profiles and decisions are saved in one project. The app shows completed steps and missing prerequisites. Pause while prints dry and resume the same project. The history helps you repeat the work and understand what has changed.

Checks of both the profile and the print

InkProf checks the profile numerically and can compare its predictions with a separately printed and measured target. Colour differences, gradients and gamut views help you assess the result and identify areas that need additional colour samples.

Benefits for photographers, print businesses and clients

Documented conditions and recurring checks support consistent prints. Finding errors early can reduce reprints, working time, paper and ink use. The ICC profile and report provide a shared basis for assessing and explaining the delivery.

Open methods and clear conditions

The source code is open and the reports document how results were obtained. Profile capabilities depend on what the printer, paper and ink can reproduce. This handbook describes 1.1.0; app buttons and field names are English.

> InkProf brings profiling, checks and documentation together in one project. Print the colour targets using a separate printing application.

PREPARATION AND INSTALLATION

What you need

Install the dependencies first, then InkProf. These installation steps apply to macOS, the tested environment.

Hardware and printing software

MATLAB needs a licence; base MATLAB is sufficient. Python 3.11-3.13 is supported for analysis and reports; the examples use tested version 3.13. Adapt python3.13 and python@3.13 to your installed version. Install ArgyllCMS too (3.5.0 used in development). Git is needed for cloning.

1. Install MATLAB, Python and ArgyllCMS

MATLAB requires its own licence; InkProf was tested with R2025b and no mandatory add-on toolboxes. Install Python 3.11-3.13 (tested with 3.13) and ArgyllCMS (3.5.0 was used in development). Git is needed for the clone command below. On macOS, Python, Git and ArgyllCMS can be installed using Homebrew; with Homebrew installed: brew install python@3.13 git argyll-cms.

2. Download InkProf v1.1.1

Download the v1.1.1 source ZIP, extract it and open the folder in Terminal. Alternatively run: git clone --branch v1.1.1 https://github.com/chto0703009/InkProf-public.git
Then enter the folder: cd InkProf-public

3. Install Python packages in the InkProf folder

Run in Terminal:
python3.13 -m venv .venv
.venv/bin/python -m pip install -r requirements-report.txt
The dependency chain includes measurement, colour calculation and PDF packages. InkProf discovers the local .venv environment.

4. Check and start in MATLAB

Set the InkProf folder as Current Folder. Run:
paths = setupInkProf(CheckPython=true);
startInkProfApp();
Run setupInkProf in each new MATLAB session. ArgyllCMS is normally discovered automatically on macOS. If needed, specify its tools folder using setupInkProf(ArgyllBin="/path/to/Argyll/bin",CheckPython=true).

Windows and Linux have not been verified for the complete workflow. Close other instrument applications before measuring and follow the calibration prompts when the instrument starts.

InkProf source and published releases

THE WORKFLOW AT A GLANCE

From start to finished profile

Six overview moments; app row numbers are separate. From planning to a verified profile and a documented decision.



After refinement: rebuild and verify with a fresh print (moments 4-5).

1. Plan project and base target

Record printing conditions and acceptance criteria. Create or import the target and save TIFF16.

2. Print, dry and measure

Keep target RGB unchanged. Use matching TI2, calibrate the instrument and measure every page.

3. Review and lock inputs

Select the measurement revision, correct suspect readings and accept the inputs before B1. Record the OBA decision.

4. Build and check ICC

Choose the profiling route, build the profile and inspect numerical checks and gradients.

5. Verify the final profile in print

Create a fresh C2 print, measure it and review C3. Refine when warranted, then repeat the build and checks.

6. Approve and save delivery

Approve for the stated use. Save ICC, certificate and the complete delivery folder.

Verifying an existing ICC has a separate workflow. Approval without a check print is a specific, numerically limited alternative and is described as such in the certificate.

OVERVIEW

Find your way through the workflow

The row number on the left and the internal step identifier are different things. This handbook uses row numbers to help you find the step in the app.

Row	App label	Purpose / further reading
1-2	Create / import RGB target; Save TIFF16	Base target and print file. Page 12.
3-4	Measure / select measurement revision; Review and accept measurement	Measure, choose a saved revision and review it. Pages 13-15.
5-7	B1. Lock profiling inputs; B2. Create recipe; B3. Build ICC	Lock inputs, choose a recipe and build a profile. Page 16.
8	C1. Fit, Grid and C1	Numerical checks. Page 18.
9-10	C2. Save verification target; Measure / select verification revision	Print and measure a verification target. Page 19.
11-12	C3. Compare model and print; Review feedback	Assess the actual print and feedback. Page 19.
13-14	Approve for intended use; Save ICC and measurement certificate	Decision and delivery after verification. Page 20.
15-17	Save refinement TIFF16; Measure refinement; Build next iteration	New colour patches and a new profile. Pages 21-23.
18	Compare with previous iteration	Compare profiles on a common basis. Page 24.
19	Save ICC + measurement certificate without print verification	Limited decision without a new verification print. Page 29.

There is more than one route

B2 is needed for the manual profiling route. The profile comparison on row 18 supports your decision. Row 19 is a separate option when the current iteration has been checked numerically but has not been verified by a separate print and measurement.

Further reading: pages 25-69

25: FWA. 26-34: profile routes, decisions, delivery and glossary. 35-37: smoothing, gradients and remeasurement. 38-41: verification and troubleshooting. 42-43: candidate comparison and photographic gradients. 44: certificate evidence. 45-47: rendering intents and ICC. 48-54: full workflow. 55-59: refinement and saved targets. 60: soft proof. 61: Python. 62: ICC grid. 63: next row and appendices. 64-67: target methods. 68-69: Jacobian. 70: later OBA/FWA choice. 71-72: measurement checkpoints, resume and RGB controls (v1.1.1).

STATUS AND NEXT ACTION

Reading the status

Status applies to an individual project step. It shows what the app has recorded and what you can do next.

Ready

Required inputs are available. Select the step and Run selected step to start. Example: creating a target can make the Save TIFF16 step Ready.

Complete

The step’s result or decision is saved and the app’s requirements for that step are met. This does not itself mean approved print quality. Example: a completed profile calculation makes the build step Complete. Print quality still needs assessment in the verification steps.

Locked

The step cannot start yet. Read its description and first complete the prerequisite steps. Example: profile building can remain Locked until the measurement is reviewed and required inputs or recipes are available.

Stale

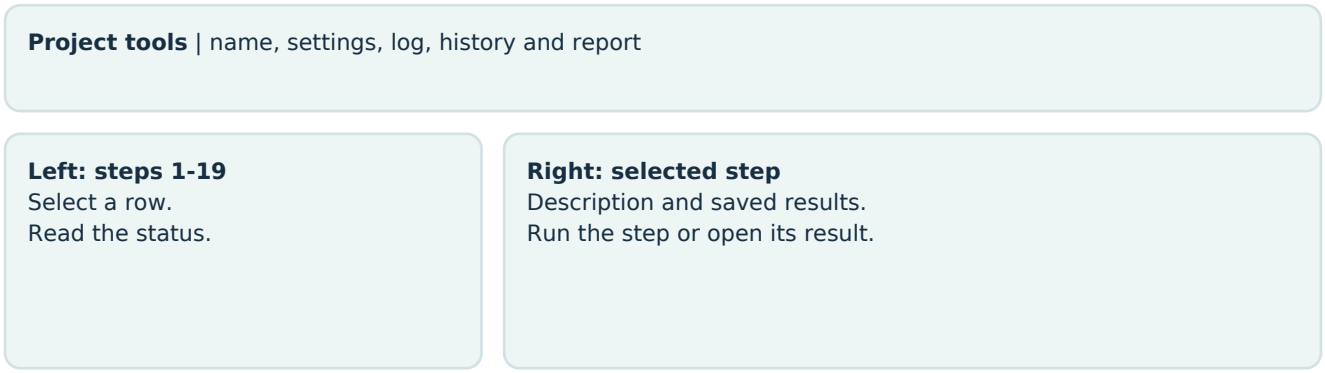
An earlier result needs updating because its inputs or settings have changed. Repeat the affected steps. Example: changing the recipe can make the earlier profile Stale. Rebuild it and update the affected checks.

Use the dialog’s intended save or acceptance button. Simply opening and closing the dialog does not complete the step.

THE APP HUB

The main window

First select a row in the table. Read the description and saved results before choosing Run selected step.



Schematic guide, not a screenshot. The app also displays status and activity.

Projects and resuming work

New project creates a project. Open project opens an existing one, including after the app has been closed while prints dry. Project details collects user, material and settings information. The project name and current path appear in the main window.

Running steps and viewing results

Run selected step opens the next operation or dialog. Open selected step results shows what has already been created. Copy TIFF16 for printing… copies a saved target file for separate printing. Read the result path to make sure you use the correct target and iteration.

History and checks

Iteration history shows the iterations. Open results log opens the work log. Verify project checks the registered files and project integrity. Refresh reloads the current state; it does not replace Save, Accept or explicit approval in a dialog.

Reports, views and closing

Open report opens an available report; the label may refer to the certificate. View 3D shows available colour points. View gamut shows a separate ICC-derived 3D surface. Close app provides an explicit way to exit. First finish any active dialog using its intended save or cancel button.

EXAMPLE IMAGE | THE APP HUB

Example: the main window

A newly created example project. The first step is Ready; later steps wait for their input data.

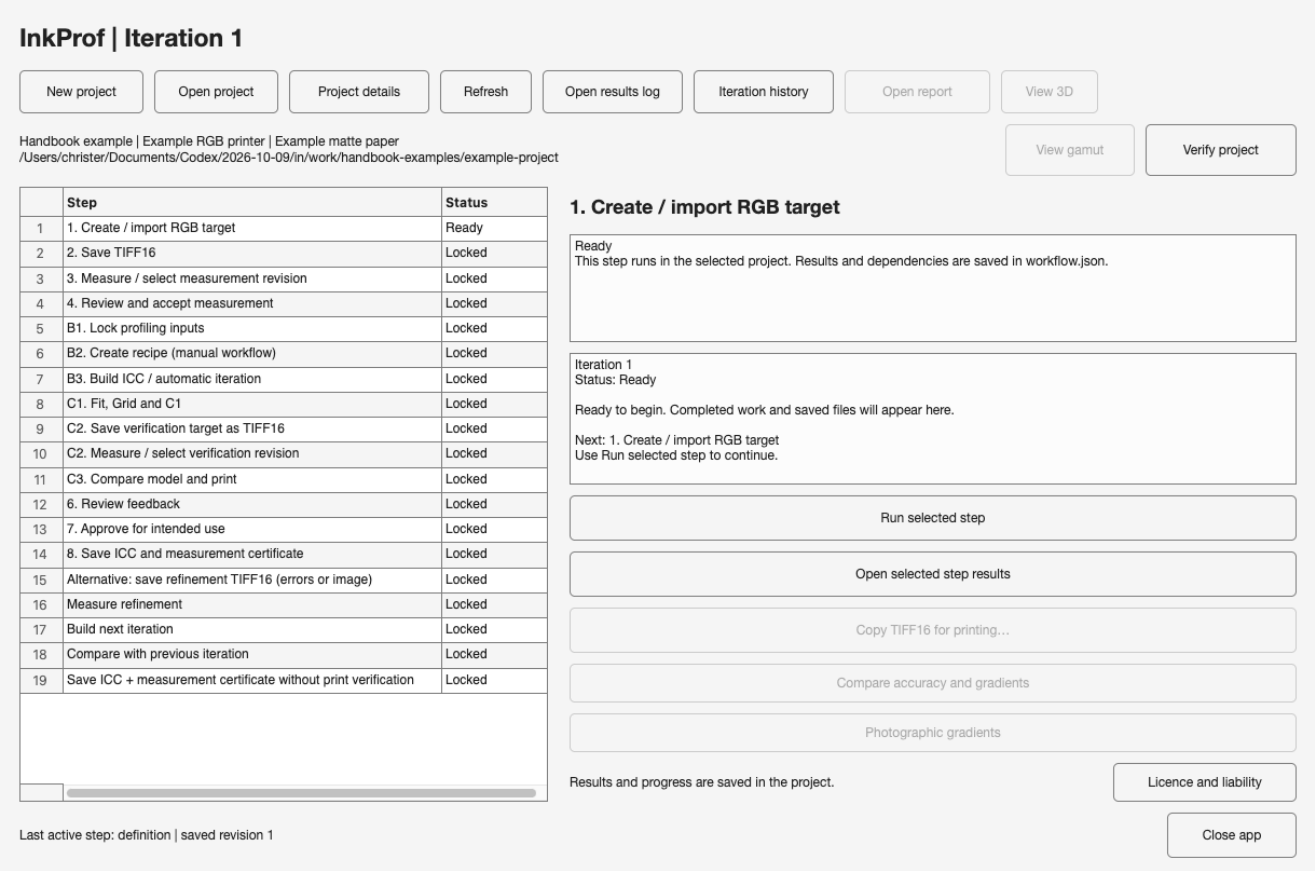


Figure 1. InkProf’s main window with fictional project details. Available buttons and status depend on the project.

Read from left to right

Choose a step in the left table. The right pane shows its description, status and saved results. Run selected step opens the operation; Open selected step results opens existing saved results.

Tools around the workflow

The top row contains project details, log, history and report controls. View gamut sits above the workspace and becomes available when the project has a suitable ICC profile.

NEW PROJECT / OPEN PROJECT

The project remembers your work

A project is a separate folder that keeps settings, files, measurements and history together.

Create and enter details

Choose New project and enter the name, user and project information. Project details has four tabs: Project and materials, Printing settings, Profiling and Target paper. Use Save to save. Cancel leaves the dialog without saving changes.

Project and materials

Enter Project name, User, Printer, Paper name, Paper finish, Ink / ink set and Ink type (dye/pigment/mixed/unknown). Use product names precise enough to reproduce the combination. Paper finish includes Glossy and Matte. This tab also contains the FWA / OBA optical-brightener option, explained on the next page.

Printing settings

Fill in Driver media setting, Printing application, Driver / RIP, Print quality, Colour management, Drying time (hours), Printer coating and Coating settings. Use Printer settings to describe additional choices. The workflow reads the drying time here; recording a duration does not itself establish that the sheet has dried.

Correct details or start again?

Correct typing errors in Project details. The recipe reads shared information here, and shared project information is edited here. A later FWA choice also updates the project. If the printer, paper, ink or printing conditions will actually form a different combination, a new project is clearer. Follow the app requirements to repeat steps when inputs change.

Rename and move

Changing the project name in the app also renames its folder. If the folder is renamed outside the app, the app can detect the difference and ask you to confirm the intended name. Move the entire project folder to another computer and open it there. Do not copy only workflow.json or a single ICC file if you want to preserve the complete workflow.

What is saved?

Project JSON files track identity, settings, steps and iterations. The results log is available as readable text and JSONL, with one log entry per line. Raw data and generated results are linked to each operation. Checksums help the app detect changed inputs.

Pausing while prints dry: finish saving the current step, close the app and reopen the same project when the sheets are ready. Do not create a new project to continue the same measurement session.

EXAMPLE IMAGE | PROJECT DETAILS

Example: project details

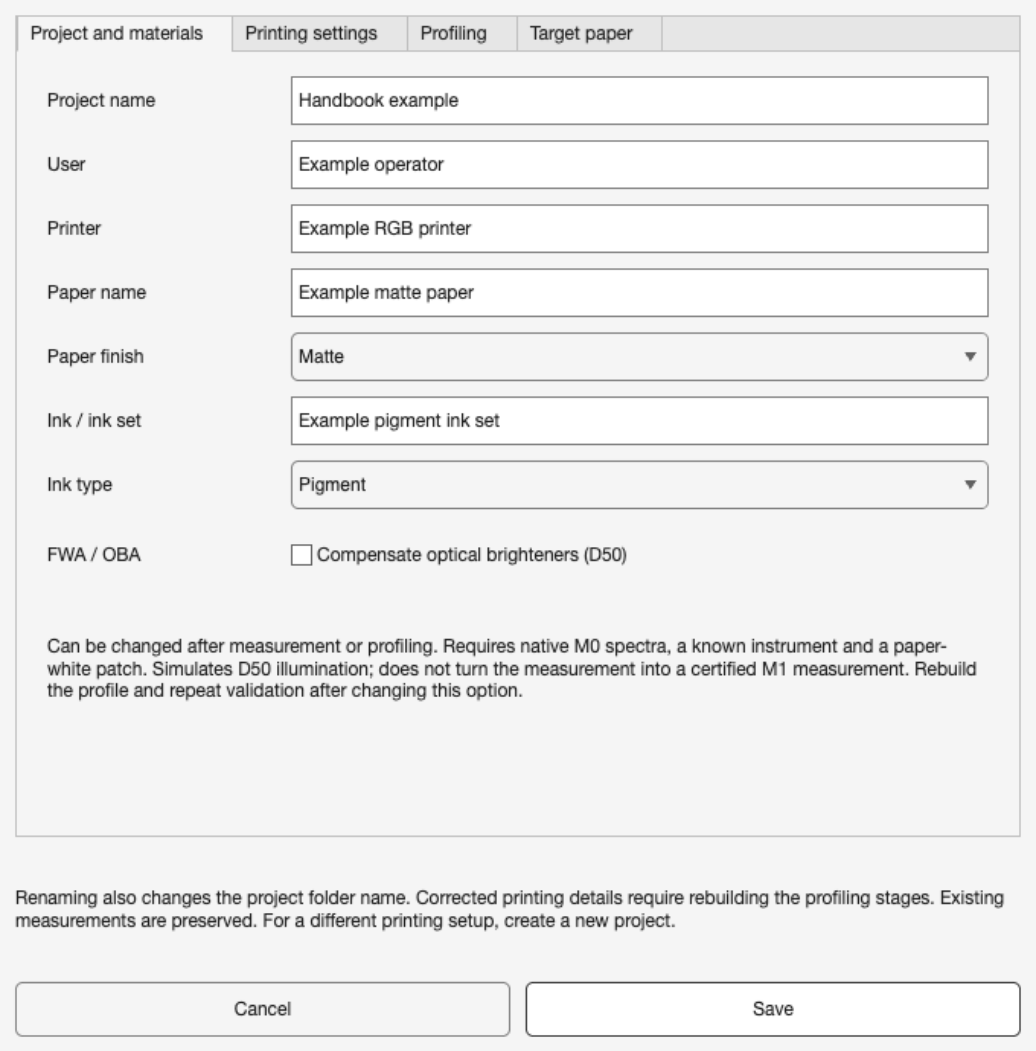


Figure 2. Project and materials in Project details. All product and user details shown are examples.

Document the combination

Record the printer, paper, finish and ink so that conditions can be reproduced. Printing settings collects driver/RIP, media setting and other print conditions.

Save what you actually use

Matte and Pigment illustrate the fields. Choose details and the FWA/OBA setting for your material and measurement evidence. Save records changes; Cancel leaves the dialog unchanged.

PROJECT DETAILS

Profile and measurement conditions

Profiling

Profile name and Description identify the profile. Colour data selects Spectra (D50 / 2 degrees) or Stored XYZ. Inverse table (B2A) selects High or Medium. Delivered ICC versions offers v2, v4.4 and Both. InkProf can import RGB printer profiles in ICC v2 and v4 and deliver v2 and v4.4. Older projects default to v2. The internal v2 profile is retained for Argyll calculations.

D50 and D65

D50 is the reference illuminant for spectral profiling in the app, together with the two-degree standard observer. D65 is another daylight reference. Assess colour under documented viewing conditions and do not mix comparisons using different illuminants. The current Profiling tab has no separate D65 option; Spectra explicitly specifies D50.

Optical brighteners: FWA / OBA

FWA (Fluorescent Whitening Agents) and OBA (Optical Brightening Agents) are two names for optical brighteners in paper. OBC (Optical Brightener Compensation) means compensation for their effect: the same phenomenon, but a process rather than a substance. Argyll calls it FWA compensation; X-Rite uses OBC. The algorithms are not necessarily identical. Compensate optical brighteners (D50) selects InkProf's compensation. The supported M0 route models D50 and does not become a physical M1 measurement.

Change FWA after measurement

Changing only FWA/OBA preserves locked B1 but makes B2 and later profile results stale. Raw measurements remain preserved. Actual changes to materials or printing conditions also invalidate B1. New recipes support FWA before Argyll pre-smoothing and final smoothing; see the M0 white-reference section. See page 25.

Black point compensation: BPC

Black point compensation maps the black point between colour spaces. It is off in the controlled comparisons described here so that the calculation conditions remain explicit. BPC off describes a condition; it is not an error message. See page 46 for version choice and BPC in perceptual production rendering.

Computer and instrument

MATLAB provides the app interface. Python, ArgyllCMS and LittleCMS are used for underlying calculations and tools. Tool paths depend on the computer and must work on the current machine. Check the installation even when the project folder is complete. The program has been tested on macOS. Windows and Linux are not verified environments for the complete InkProf workflow.

EXAMPLE IMAGE | PROFILING

Example: profiling settings

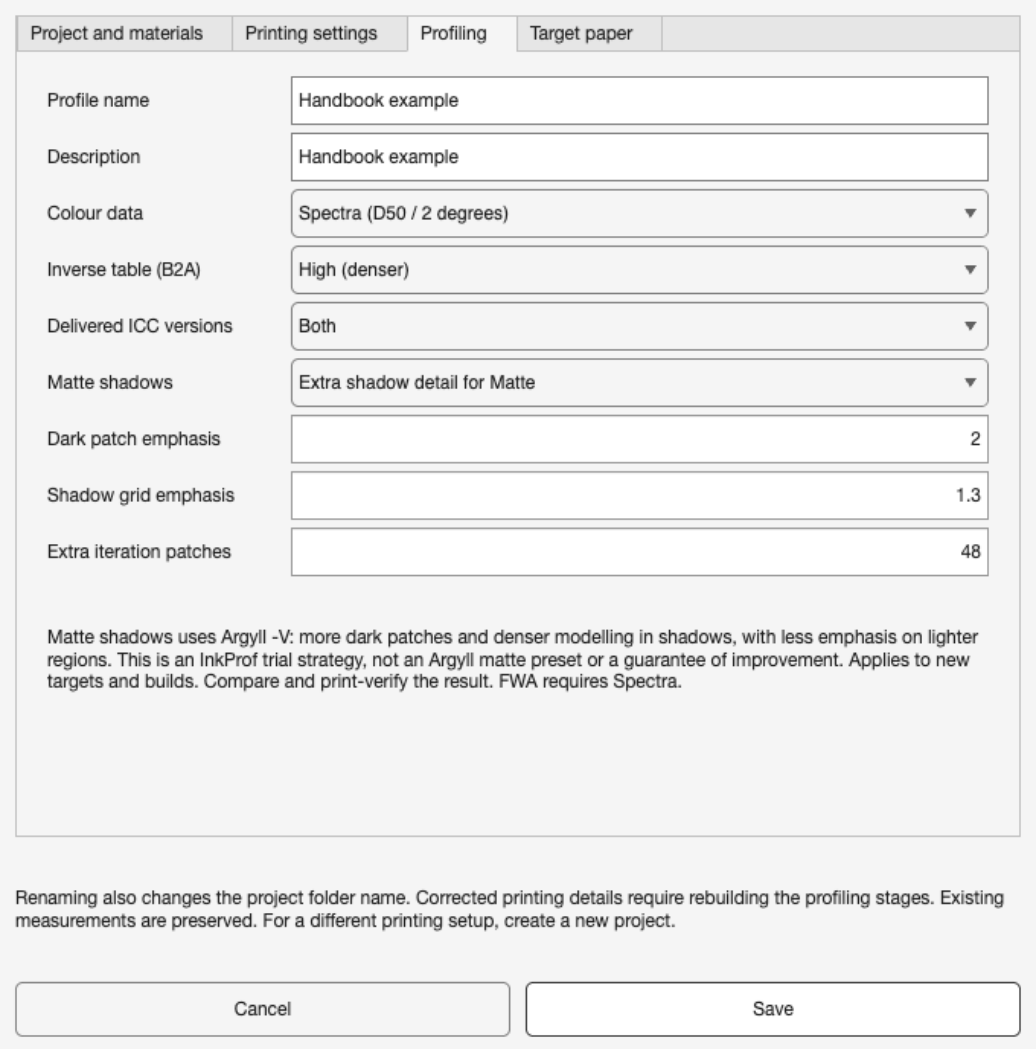


Figure 3. The project definition’s Profiling tab with the new version choice. Both is selected as an example; these values are not a general recommendation.

What the fields show

Profile name and Description identify the profile. Colour data selects spectra or stored XYZ. Inverse table (B2A) specifies table quality. Delivered ICC versions shows Both in this example: delivery of v2 and v4.4.

Shadows and colour samples

Matte shadows and the next three fields describe additional shadow detail in targets and profiling. Review the print result before deciding whether these settings help your combination.

ROWS 1-2 / PAPER DIALOG

Base target and TIFF16

A base target defines the RGB values the printer should reproduce. The printed and measured version becomes the input for profiling.

Row 1: create or import

Run Create / import RGB target. Generate a target or import a compatible RGB definition. Preserve patch count and identity through layout, printing and measurement. All methods and grid settings are explained on pages 64-67.

Row 2: save the print file

Run Save TIFF16 and choose a layout. The app saves a 16-bit TIFF with matching target and layout information. Check page count, patch count and the recorded file path in the result. TIFF16 is the print deliverable; InkProf does not send the job to the printer.

Target paper and the measurement sled

In Project details, Maximum measurement sweep (mm) sets the greatest width of a sweep. Maximum target length (mm) limits the other target dimension. Roll width (mm) sets the roll width. Limits include margins and are saved in project JSON. Adapt them to your equipment; they are not universal instrument dimensions.

Editable paper suggestions

Paper planning compares A5 to A3+, cut pieces and roll feed. 1/2 A4 (2 pieces per sheet) means a half-A4 measurement piece, not a split TIFF. Whole sheet uses the entire sheet. Dimensions and areas display one decimal; calculations retain precision. Check that the piece fits the printer and measurement equipment.

Print and allow to dry

Copy the target with Copy TIFF16 for printing… if needed. Print at 100% size without automatic fit-to-page scaling. Use the documented materials and driver settings. For an unconverted base target, no additional profile conversion should change the intended RGB values. Save the print settings and follow the project drying time.

Find the right row when measuring

Use row and position labels to orient the sheet. Labels on both sides make scanning easier. A previously exported TIFF does not change when the program code is updated; inspect the actual file you will print.

Why this step matters: incorrect scale, colour management or paper settings can produce measurements that do not describe the printer combination you intend to profile.

ROW 3 / MEASURE CHART

The measurement window

Measurements use a spectrometer supported by the measurement workflow. Instrument calibration and the correct target-file association are essential.

Before starting

Choose Measure / select measurement revision. In the measurement window, check Chart (TI2), Scan mode, Scan tolerance, Instrument port and Measurement condition. The target file must match the printed sheet. Instrument port 0 means the default selection.

Start and calibrate

Start measurement starts the instrument workflow. Calibrate when requested. At page changes, instructions show the page, row number and FORWARD or REVERSE, consistently with the row prompts. Position the correct page and use Page loaded. Instrument messages appear in the log.

Scan and retry

Single direction, Alternate rows and Forward + reverse average describe different scanning patterns. Follow the chosen mode. Reread / retry repeats a reading. Accept reading accepts it when requested. Previous row, Next row and Next unread help you navigate. Use the sheet labels and status instructions, not only letter codes in the log.

Finish with a saved result

Save and finish ends the measurement and saves a new revision. The overview of all patches may open after Save; this is intended review and does not start remeasurement. Earlier revisions remain available. If no successful save was recorded, no automatic import occurs; read the log before continuing.

Measurement conditions and instrument identity

The window distinguishes conditions including M0 for i1Pro 2 without a UV filter and an unspecified instrument-default mode. Do not specify M1 or M2 unless the instrument workflow actually provides it. Model and serial number are read from instrument output when available; do not replace unknown details with assumptions.

If the first connection fails

For some early startup failures, the measurement workflow can make limited retry attempts. This is not the same as a full hardware reset. If an error persists, cancel cleanly, read the log and check the connection and whether another program is using the instrument. Then start measurement again.

New pages have separate RGB print controls. InkProf checks them before strip measurement and stops large differences from an approved reference. Check active Photoshop and driver settings even when using presets. See page 57.

EXAMPLE IMAGE | MEASURE CHART

Example: measurement

The measurement window before starting and the instrument positioned on the printed target. The window screenshot is an example without a completed physical measurement.

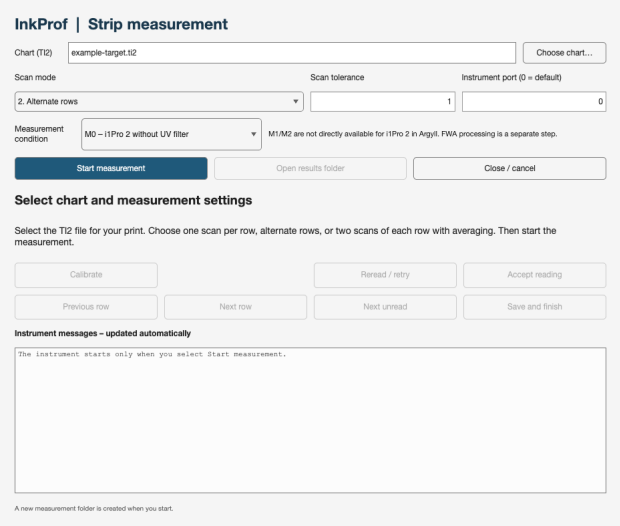


Figure 4a. The measurement window with an example TI2 and alternating row scans. The instrument starts only after you choose Start measurement.



Figure 4b. The spectrometer positioned over the printed colour target. Use the sheet row markings and follow the scan direction shown in the measurement window.

Choose the target and follow instrument messages

Use the TI2 matching your printed target. Review Scan mode and Measurement condition before starting. Calibration, remeasurement and acceptance buttons activate when needed. Instrument messages tell you what to do; Save and finish saves the completed result.

The instrument on the sheet

Lay the sheet flat and keep the instrument correctly positioned during the scan. Start and finish as instructed for the row. The photograph illustrates positioning; associate the measurement with the correct page and the TI2 matching the print.

ROWS 3-4 / SAVED REVISIONS

Revisions, review and remeasurement

A measurement displayed on screen is not automatically the revision used for profiling.

Saved revisions: select the saved input

After measurement or import, the revision selector lists available saved measurements. Identify the correct session by date, target and coverage. Use revision makes the selected revision the workflow input. Check which revision is now shown as being used.

Row 4: review and accept

Review and accept measurement opens the measurement overview and associated assessment. Review missing values, obvious deviations and patches needing attention. Enter relevant notes and finish using the explicit acceptance action. Closing the preview is not approval.

When a patch needs remeasurement

Remeasure patch rereads one patch. Remeasure row lets you choose a page and printed row and retains the original scan mode, including forward/reverse. Review changes before Accept row replacement. Discard / Close leaves the row unreplaced. An accepted row is saved as a new revision; original and raw readings remain preserved. See page 37.

Absolute colour and change are different

Measured Lab describes the colour itself through L*, a* and b*. The change between two measurements describes how those values changed. ΔE00 summarises the colour difference. Read the label before interpreting a number as a new absolute value, a component change or an overall colour difference.

Changed measurements affect the workflow

Using a new revision can make locked inputs, the profile, checks and approval stale. This is intentional: the finished profile must be traceable to the measurement actually used. Earlier revisions provide history but do not replace updating dependent steps.

Always check three things: has the remeasurement been saved, is the new revision selected, and has the review been accepted? Refresh can show the current state but does not perform these actions.

ROWS 5-7 / B1, B2 AND B3

Build an ICC candidate

An ICC profile describes the relationship between printer RGB values and the colours reproduced. The candidate then needs to be checked.

Row 5: B1. Lock profiling inputs

B1 locks the selected, reviewed measurement together with the necessary target and project information. This creates a traceable starting point so that profile building does not accidentally mix revisions. Repeat the step when the app indicates that the inputs have become stale.

What is a RoleFile?

A role file describes how patches should be used, for example for training, control or reserved assessment. If the dialog offers No separate role file and you do not have a purpose-made role file, choose that option. Do not invent roles or select an unrelated file. For app-generated refinement targets, role associations are part of the saved evidence.

Row 6: B2. Create recipe

B2 is used by the manual route. Measurement data smoothing offers Argyll only (raw measurements) and Argyll colprof pre-smoothing (experimental). Assumed noise (%) belongs to pre-smoothing; Smoothing at ICC build controls final colprof -r (empty = 0.5%). See pages 26 and 35. Perceptual compression (%) is explained on page 45.

Row 7: B3. Build ICC / automatic iteration

Manual uses the B2 recipe. Automatic creates its own recipes and does not use B2's gradient checkbox. After a successful manual build, the RGB gradient window can open if selected in B2 and pre-regularization is enabled. Saving B2 does not open it. Complete means calculation finished, not approved print quality. Automatic carries forward B2 perceptual compression; preprocessing and the gradient checkbox remain part of the Manual route.

What is saved when it finishes?

The project retains the v2 ICC candidate with its locked inputs and profile job. Selecting v4.4 or Both also produces a separate v4.4 profile and conversion report. Colorimetric tables are preserved; perceptual and saturation tables are remapped. Assess v4.4 in the application and rendering intent used for production.

Why is creating an ICC file not enough?

A successful build shows that a profile could be calculated. It does not by itself show how a new print will turn out. Numerical checks follow, and the full verification route also requires a separate verification print and measurement.

EXAMPLE IMAGE | PROFILING RECIPE

Example: profiling

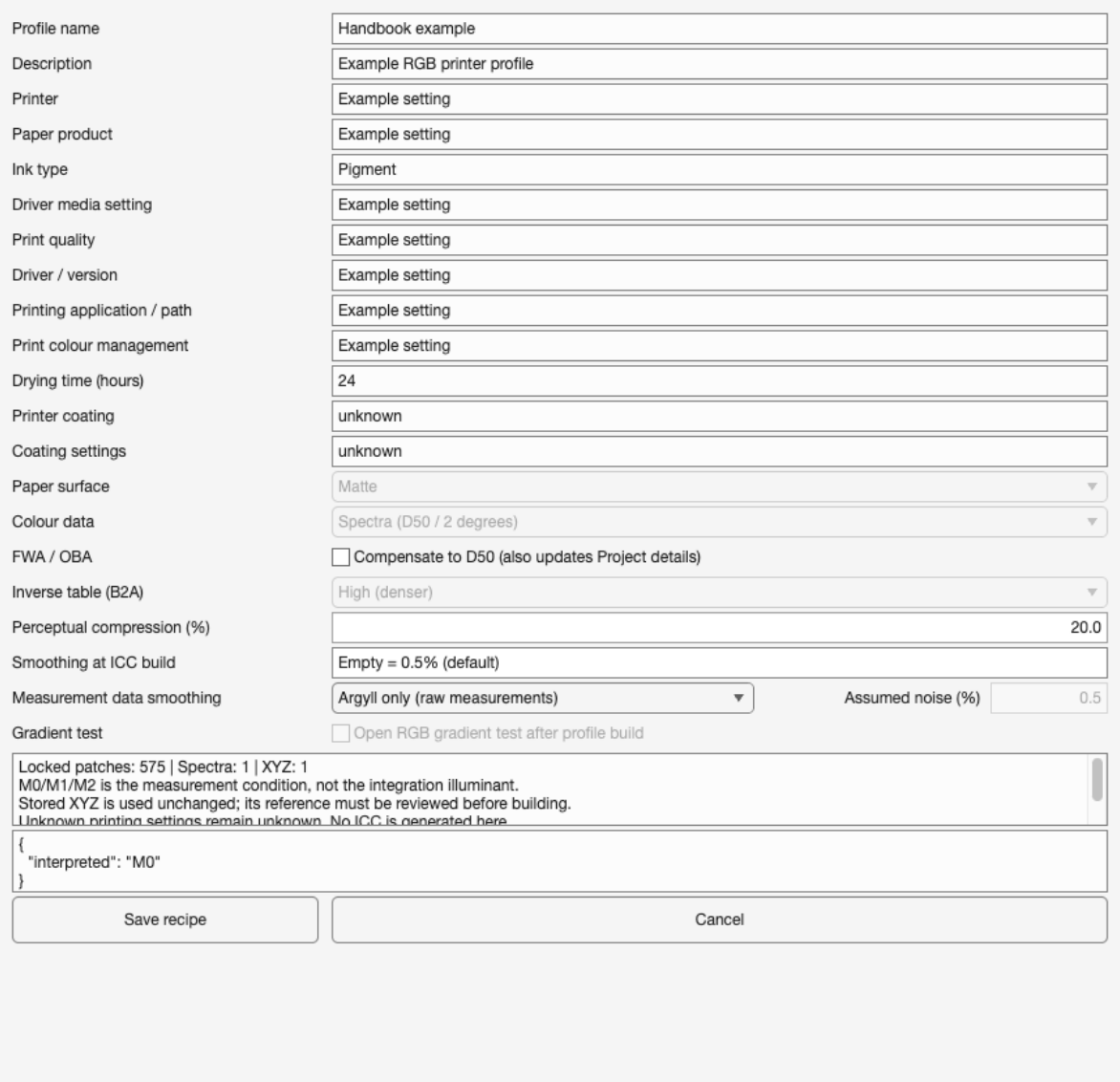


Figure 5. Profiling recipe in the manual workflow. The 575 patches, materials and settings are examples; no ICC was built for this image.

From recipe to ICC profile

Review colour data, table quality and optional smoothing. Project print details are locked in the recipe. Save recipe records settings; B3 then builds the profile. The recipe window itself does not create an ICC.

ROW 8 / C1. FIT, GRID AND C1

Understand the numerical checks

These checks examine the calculated model and its tables before you assess an actual verification print.

Item	What it helps you understand
Fit	How closely the model agrees with the measurement data used for comparison. Training data are not an independent verification print.
Grid	Profile behaviour on a chosen calculation grid. Denser sampling can reveal detail but remains a numerical investigation.
Forward and inverse	Forward calculates colour from RGB. Inverse calculates RGB for a desired colour. Results from different engines can be compared.
Local probes and ramps	How RGB or colour changes with small perturbations and along transitions. Strong changes need to be interpreted in context.
Mean, median, P95, max	Average, middle value, 95th percentile and greatest deviation. Read both typical behaviour and individual extremes.

Distinguish the error measures

ΔE00 measures colour difference using CIEDE2000. RGB deviation in percentage points describes a difference in device values. A small difference between two calculation engines can coexist with a larger difference between desired and printed colour. These measures answer different questions.

Is a large step a discontinuity?

A steep transition or interpolation knot is not itself proof of a discontinuity. Investigate several step sizes and neighbouring samples. Finite numerical sampling cannot prove continuity throughout the colour space.

Next action

Open the saved check results. Read any warnings and their supporting evidence. Continue with C2 to check a print, or later use the explicitly limited numerical delivery route if that is your documented decision.

A small numerical deviation is not the same as a small print deviation. Always distinguish calculated results from what was actually measured on paper.

ROWS 9-12 / C2, C3 AND FEEDBACK

Check the actual print

Row 9: save the verification target

C2 creates a verification target for the current ICC. Lab references are converted once to printer RGB with absolute colorimetry and no BPC. Preparation displays activity and elapsed time. Save TIFF16, print separately and allow it to dry. C2 contains colours, neutrals, challenges and repeats, not only problem regions.

Avoid double profile conversion

New C2 files contain the printer profile as an embedded tag. The tag does not change RGB values or mean the profile should be applied again. Keep it without conversion. Older files may be untagged. Printer Manages Colors or a greyed-out Off control does not prove the complete print chain has no extra conversion. See page 38.

Row 10: measure and select a revision

Measure / select verification revision uses the verification target definition. Measure, save and select the correct revision just as for the base measurement. A target from an earlier ICC candidate does not automatically verify the new candidate.

Row 11: C3. Compare model and print

Measured vs desired compares print with desired colour. Model vs measurement compares ICC prediction with print. Both are colour differences, not absolute colour values. Desired Lab and Measured Lab contain L*, a*, b*. Largest errors appear first. C3 may warn about suspected double colour management; this is a hypothesis, not proof. See page 39.

Row 12: Review feedback

Open and read the feedback, then document your assessment using the intended completion action. A remaining Ready status may mean you only viewed the result. Go back and check that the assessment was actually saved.

What does not approved mean?

A statement that colour management or acceptance criteria are unverified means the result does not yet have a documented approval decision. It does not by itself mean measurement failed. Read the available evidence and outstanding requirements before choosing the intended use.

A high Measured vs desired but low Model vs measurement may suggest that the model predicts the limitation reasonably well. A large model error warrants closer review. Neither case establishes the cause without further evidence.

ROWS 13-14 / VERIFIED ROUTE

Decision, ICC and certificate

Row 13: Approve for intended use

Review current C3 results and run row 12 first. In row 13, enter the intended use and accepted limitations in Your assessment. Tick the confirmation that you reviewed the results and checked the print settings against the project. Select Approve for stated use. Row 14 includes the saved decision in the certificate. See the walkthrough on page 58.

When approval becomes stale

Changes to the ICC, inputs or checks may mark approval as stale. Read which dependency changed and update the required steps. An old decision must not simply carry over to a new profile.

Row 14: choose what to save

Save ICC and measurement certificate follows Delivered ICC versions: v2, v4.4 or Both. Both saves v2 at the chosen filename and a sibling <name>-v4.4.icc. The project working profile is retained. Changing only the version choice requires a new export, not new measurement or profiling.

Profile names in files and image applications

The profile name comes from Profile name in Project details without an additional prompt. It identifies the ICC filename and internal description. The suggested report name is <profile name>_iter-<number>_<last eight UUID characters>.pdf. The full UUID remains in report data for traceability. Select Save delivery elsewhere in step 14. The save dialog starts in Downloads; the main window is temporarily hidden until Save or Cancel.

Certificate contents

The measurement certificate describes the project, materials, printing conditions, profile and check results. It should identify the iteration, the measurements supporting it and how FWA was handled. Instrument identity is reported when available. Report colour patches are screen previews, not substitutes for the original print. For v4.4 delivery, the report distinguishes the checked v2 from the converted v4.4 copy; it does not certify a separate verification print for its perceptual rendering.The certificate is in English. See page 63 for the document structure.

Where are reports saved?

The project retains its results. For external delivery, choose the certificate name and location. A folder named after the PDF without .pdf contains the ICC, PDF, HTML and supporting evidence. Both places both ICC versions in this folder. If the folder exists, a number is added to it and the reports. Move the whole folder for HTML; the PDF can be opened separately. Open report opens the project report.

See page 63 for the main certificate, Appendices A/B and separate measurement appendix. It records scope and decisions; it is not accredited certification.

15 | REFINEMENT TOOLS

Select new colour patches

Refinement tools groups image, gamut, shell, error and grid refinement. See page 56 for gamut/shell and page 62 for the ICC grid trial.

From verification errors

From verification errors uses available verification and feedback. Describe why you want to continue. A request for a reason does not mean the app has proved another iteration would be useless; it documents your decision.

From an image

From image extracts representative colours from an RGB image. Choose an image with an embedded ICC profile. If it has none, the app warns you and offers Use sRGB, Choose another image or Cancel. sRGB is then an explicit, recorded assumption. The original image is not changed.

Image region and patch count

The dialog lets you limit the image region and number of new patches. A minimum RGB distance reduces close duplicates. A neighbourhood radius can add samples around selected colours; a zero radius gives image colours without these neighbours. An image containing many similar pixels does not necessarily need as many separate patches.

Review image suggestions

The selection window shows calculated measures and colour patches. The default filter is an estimated local fit deviation ΔE00 greater than 5. You can change the threshold. Only visible, selected suggestions proceed. If none appear, review the threshold and chosen measure; it is not automatically proof of perfect colour.

What does estimated error mean?

Estimated local fit uses nearby training measurements to describe local model fit. It is not a new measurement of the printed image. Model mapping ΔE00 is a different measure: the deviation in the profile’s calculated reproduction of the desired image colour. Missing evidence for a local estimate must not be interpreted as zero error.

Save the selection and choose paper

Confirm the selected patches and review the suggested paper layout. Use saved image patches can reuse a saved selection. The app saves inputs and choices for traceability. Read the patch count and TIFF path in the result before printing.

ROW 15 / ARGYLL OR INKPROF

Choose the error-driven refinement method

Choose the method in row 15

Choose From verification errors, then 1. Argyll only or 2. Current InkProf (Jacobian sampling). Current C3 verification and feedback are required. Argyll only is the default. This selector is for additional patches, not B3.

Argyll only

The checked ICC preconditions Argyll targen to select new device-RGB patches, without InkProf Jacobian sampling or pre-regularization. The continuation builds a high-quality colprof candidate with -r 1.0.

Current InkProf (Jacobian sampling)

C3 residuals and Jacobian-guided patch selection are used, followed by comparison of ICC candidates using development measurements. Both methods build the ICC with Argyll colprof; they differ in patch selection and continuation settings.

Saved choice and verification

The method is saved in proposal.json as refinementMode and passed to row 17 after the new target is printed and measured. Choose another method when creating a later error-driven refinement; existing targets and measurements are not rewritten. Image-guided refinement uses a separate path. Numerical B3 fit checks are not independent print verification: print and measure the new C2 target to assess the new result.

Jacobian explained

See pages 68-69 for local sensitivity, a worked example, actual candidate selection and review.

ROWS 15-17 / COMBINED TARGET

Measure the refinement and iterate

C2 is included until it is measured

When the current C2 verification has not yet been measured, its patches can be included with the image suggestions in the refinement target. Printing C2 is not enough to exclude it. A valid registered measurement for the current C2 determines whether it is still pending.

Patches have different roles

The combined target preserves C2 device values and references. Image suggestions and additional control patches are added. An image ΔE00 filter does not remove pending C2 patches. The total patch count may therefore exceed the number of selected image colours.

Row 16: Measure refinement

Print the correct TIFF16 and measure the combined target using its matching target file. Save and use the revision. When target and measurement are correctly linked, the same measurement can also provide C2 evidence and the C3 comparison for the preceding candidate. You should not need to measure the combined sheet twice.

C2 measurements in training

C2 colour, gray and challenge patches in the combined data may be used to train the next profile. Repeats and paper-white patches are handled as controls to prevent unintended double weighting. The saved target roles determine their use.

Row 17: Build next iteration

The next iteration uses the previous locked inputs and the new training patches specified by the saved continuation plan. The app builds a new ICC and performs associated checks. When calculation finishes, choose comparison, new physical verification or the limited delivery route on row 19.

An important boundary between training and verification

A C2 measurement can assess the profile that created its target. Once the same measurement is used to train the next ICC, it is no longer an independent check of that new ICC. Independent checking requires new verification evidence. Do not assume separately measured C2 data are automatically included in every later training run; read the current continuation plan.

Row 17 creates the next profile inside the project. This does not automatically mean external delivery. For verified delivery, continue through current verification and rows 13-14; for a numerically limited delivery, use row 19.

ROW 18 / COMPARE WITH PREVIOUS ITERATION

Compare two iterations

Comparison helps you see what changed and document why you continue or stop.

The right profiles without searching filenames

The workflow uses the project’s registered iterations and ICC files. Check the displayed profile names and iterations. Comparison is available when a previous and current candidate exist, normally from iteration 2.

Different measurement meshes can be compared

The profiles do not need to have been built from identical measurement grids. The tool evaluates them at common calculation points, putting a comparable question to both models. It does not make their original physical measurements identical or equally independent.

The 2D view starts at L* = 50

The flat view shows a* and b* at the selected lightness L*. Change L* to investigate lighter and darker regions, and use the slice width to control which points appear. Turning off 2D displays a rotatable 3D view.

Colours in the comparison

Blue represents the previous profile and orange the current profile in the iteration comparison. Always read the legend and profile names. A wider extent in a region means the models describe it differently; that alone is not a quality rating.

Interpret differences in context

Small profile differences may support stopping when the result meets the intended use. Larger or unexpected differences may warrant further review. Compare numerical measures, local transitions and available measurements. A numerical change does not itself show whether new prints are better.

Document the decision

Save the comparison as evidence and explain which observations mattered to your decision. When comparing error measures between iterations, identify the measurements and conditions they came from. This also makes later deterioration or divergence understandable.

A useful decision rationale is specific: what changed, what evidence was assessed and why is it sufficient for the intended use? Do not equate a larger gamut with lower measured error.

FWA / OBA WITH ARGYLL SMOOTHING

M0, optical brighteners and averaged white

What is measured, and what is estimated

Native i1Pro2 M0 spectra are measured under the instrument’s illumination. Argyll estimates their response under simulated D50 using paper white and instrument information. This is the best available M0-based estimate in this workflow, not direct UV excitation quantification or a native M1 measurement. A UV LED observation indicates brighteners but does not quantify the correction.

Several white patches

All device-white spectra are averaged by Argyll (RGB channels above 99.9%). Their count, mean spectrum and sample variation are saved. Averaging can reduce random variation; it cannot restore missing UV excitation or guarantee that weak fluorescence is resolved. White controls preserved in B1 source data can supply the reference even when excluded from fitting.

Compensation before smoothing

B2 recipes integrate and FWA-compensate spectra once using spec2cie. Both the intermediate Argyll pre-smoothing model and the final ICC read the resulting XYZ without repeating compensation. Fit and C3 use the same policy. Save a new B2 recipe; use Manual B3 for the explicit pre-smoothing choice. Automatic iteration still creates its own recipes.

If no white measurement is available

B2 offers Measure blank paper, Select saved blank-paper reference or Cancel. Use the same unprinted stock and backing; measure several clean locations with stationary i1Pro2 M0, then explicitly accept the mean. Follow calibration prompts. The instrument, wavelength grid and calibration standard must match. Required colour spectra must cover at or below 400 nm through at least 700 nm; measuring only white cannot repair missing colour spectra.

What is saved and reused

The approved reference includes source readings and checksums. A separately measured white supplies an auxiliary profile white anchor; it is not presented as a target colour or independent verification patch. Analytical comparisons retain the original target patch set. B2’s approved reference is available to automatic builds and continuation on the same stock when needed; current target whites take precedence.

Certificate and final check

The certificate records white-reference source and count separately for training and verification and explains the M0 limitation. Original spectra remain unchanged. Compensation or averaging is not proof of improved reproduction: finish with a new C2 print and C3 measurement before approving the profile. See page 70 for step-by-step later activation.

B2 / B3 / REFINEMENT SETTINGS

Choose the profiling and refinement path

B2: two current choices

Argyll only (raw measurements) disables separate pre-smoothing, but colprof still fits and smooths. Argyll colprof pre-smoothing (experimental) adds an intermediate model before the final ICC build with FWA integrated once before both passes in new recipes.

Two independent noise settings

Assumed noise (%) is first-pass colprof -r, default 0.5%. Smoothing at ICC build is final-pass -r; empty uses the documented engine default 0.5%. Neither value is an acceptance limit.

B3 Manual versus Automatic

Manual uses the saved B2 recipe. Automatic creates its own recipes without pre-smoothing and carries perceptual compression forward. Argyll mode uses high quality and -r 1.0 even with development data. InkProf mode compares -r 0.5, 1.0 and 1.5 when development data exist; otherwise it builds one engine-default candidate. See page 42.

Refinement method in row 15

Argyll only uses preconditioned targen patches without Jacobian sampling; continuation builds with -r 1.0 and no pre-smoothing. Current InkProf (Jacobian sampling) uses C3 residuals and Jacobian-guided patches. Both build ICCs with Argyll. InkProf continuation disables pre-smoothing and compares final -r candidates 0.5, 1.0 and 1.5 with development data; without such data it uses the engine-default build.

What continues into the next iteration

Row 17 follows the mode saved in the proposal and its specified combined training set, not the entire older B2 recipe. Read the selected candidate recipe and job log. Change refinement mode by creating a new proposal in row 15 and measuring its target.

Two practical routes

First profile: rows 1-7, choose B3 in row 7 Manual or Automatic, then rows 8-14 for measured delivery. Refinement: row 15 method choice, print and measure in 16, build in 17, compare in 18, then repeat 8-14 for final measured approval. Row 19 is numerical-only delivery.

Evidence belongs to a specific profile

Check iteration, ICC and measurement revision. A changed profile cannot inherit earlier physical verification or approval. Small prediction changes or numerical stopping do not prove print accuracy. Investigate Stale/Locked status and repeat dependent checks.

INTERPRETING FINAL RESULTS

Best practice: acceptance and print-system limits

A useful photographic profile may still fail to reproduce some reference colours within the selected comparison limits.

Agree on the intended use

Set the use, relevant colours, gradients, viewing conditions and acceptance criteria before judging results. Stabilise the printer and record paper, ink, preset, drying, backing and measurement conditions. Resolve acquisition errors before changing smoothing or refinement.

Keep three kinds of evidence separate

Training fit describes agreement with profiling measurements. Profile comparison describes changes between predictions. Final C2/C3 measures a new print against desired colours and current predictions. Once a measurement trains a profile it is no longer independent verification of that profile.

Stop for a documented reason

Use comparable errors and visual checks of neutrals, skin tones, shadows and gradients. No useful improvement may justify stopping refinement, but does not prove print accuracy or physical gamut limits. Record regularisation and the chosen Argyll only or InkProf/Jacobian refinement method.

Interpret remaining errors honestly

Large desired-colour errors with small prediction errors may be consistent with system limits, but first check profile identity, print conversion, measurement conditions, inversion and local model reliability. Model-reachable, outside-model-or-inversion-unresolved and unknown colours are distinct. Paper white, attainable black and chroma may limit reproduction. An unresolved inverse alone is not proof of physical impossibility.

Approval does not waive a standard requirement

Disclose unmet comparison limits. The ISO-related references shown in InkProf do not establish conformity of its custom RGB target. Where an applicable standard is required and its requirements are unmet, do not approve for that purpose. A reviewer may approve a measured photographic result for another stated use with explicit limitations.

Present the full result

Report all assessed colours and mean, median, P95 and maximum errors, with difficult colours separately identified. Neither worst patches alone nor the overall average is sufficient. Follow the final measured approval route in rows 8-14. Row 19 supplies only numerical evidence for the current profile.

FINAL PRINT VERIFICATION / ROWS 8-14

Finish with a measured and approved profile

When further refinement gives no useful improvement, finish by printing and measuring a control target for the exact profile you will deliver.

Decide when to stop

Use row 18 and comparable error measures to document why you stop. Training fit, model change and physical print accuracy are different measures. Small changes or numerical convergence alone do not prove print accuracy. This procedure does not assume automatic detection of stalled improvement.

1. Check and create the final target

After row 17, run row 8 (Fit, Grid and C1) for the selected current profile. Run row 9 (C2: save verification TIFF16). Check that the target belongs to this profile and iteration. The ICC transform has already been applied once.

2. Print without another colour conversion

Follow the accompanying target text. In Photoshop choose Discard the embedded profile (don’t color manage) when opening the target. Print at 100%, with colour management disabled in the application and colour adjustment disabled in the printer driver. Keep the documented paper, ink and printer settings and allow drying.

3. Measure and review

Run row 10 to measure the final control print. Run row 11 (C3) to compare measured print colours with desired colours and profile predictions. Run row 12 to review feedback. This final evidence must belong to the delivered profile; measurements used to train it are not independent verification of it.

4. Approve and deliver

Run row 13 (Approve intended use), enter the intended use, accepted limitations and review notes, and confirm explicitly. Run row 14 to save the ICC and measurement certificate. The certificate records actual print measurement and user approval tied to that profile. Approval is for the stated use, not automatic ISO certification.

Do not use row 19 for this route: it documents a numerical-only delivery without current print verification. If the final measurement does not meet the intended use, investigate or refine before approving.

ROW 19 / A CERTIFICATE WITH EXPLICIT SCOPE

Finish without another test print

When this option fits

You may be satisfied with a later iteration after numerical checks and comparison, while choosing not to print and measure another C2 target. Row 19 documents that decision and lets you save the ICC and measurement certificate with the correct scope.

What the user does

Make sure numerical checks exist for the current iteration. Open Save ICC + measurement certificate without print verification. Enter the intended use and rationale. Confirm the scope in the dialog and choose destinations for the delivery files.

The certificate must state this clearly

Numerically checked; this iteration has not been verified by a separate print and measurement.

Earlier measurements retain their identity

The last measurement results may be included as background. The measured iteration and profile must be identified. Older physical results do not become new measurements merely because they appear in the current certificate. A profile comparison may explain the decision to stop without serving as physical verification.

PDF and HTML

PDF provides a fixed document for handover and signature. HTML provides interactive profile viewing: a 2D slice initially at L* = 50 and an option to switch to 3D. PDF renders figures as still images. Move or send the entire HTML report folder so its supporting files are included.

Signature and legal appendix

Results, scope and decision precede the signing section. Legal text follows in an appendix. It covers matters including information supplied by the client, responsibility for client-printed targets and limitations of the printer, paper and ink. Read the appendix together with the documented scope when handing over the certificate.

Appendices

Row 19 also uses the documented structure, with explicit numerical scope and historical evidence. See page 63.

The document is still called a measurement certificate. Its title must not obscure the difference between measured print verification and an iteration that has only been checked numerically.

STATUS, ACTIVITY AND RECOVERY

When something seems to stop

What you see	What to do
Ready after closing a window	Check that you actually used Save, Use revision or the acceptance action. Opening a preview is not enough.
Running / calculation in progress	Read the active operation and elapsed time. Let it finish. Do not open the same operation repeatedly.
Waiting for a dialog	Check whether a file chooser, confirmation or measurement window needs your response. Follow the status line.
Stale: outdated inputs	Open the results and identify changed inputs. Repeat the necessary checks and make a current decision.
Cannot find a new TIFF	Read Saved results for the correct iteration and step. Use Open selected step results or Copy TIFF16 for printing.…
Project integrity check failed	Read which files are identified and open the results log. Do not attempt recovery by arbitrarily editing JSON or deleting integrity records.
Results exist but the list is old	Press Refresh after the current operation finishes. Then check step status and the revisions being used.

Visible activity in later steps

Longer calculations display operation and elapsed time, including C2 colour and paper preparation. TIFF generation can show a clock and Cancel. Time is elapsed time, not a forecast. Long TIFF footer summaries are shortened when needed; full filenames and metadata remain preserved.

The measurement window has its own signals

Calibration, row reading and instrument questions appear in the measurement status and log. The program may be waiting for a physical action. Distinguish this from profile and report calculations.

Reporting a problem

State the step, iteration, what you clicked and the exact error text. Include the relevant part of the results log. This helps identify the cause without guessing or changing measurement data.

REFERENCES AND TECHNICAL BASIS

Glossary, sources and version

Term	Meaning
ICC	International Color Consortium; here, a profile describing colour conversion for a device or colour space.
CIELAB / L*, a*, b*	Colour space from the Commission Internationale de l'Éclairage. L* describes lightness; a* and b* describe colour directions.
ΔE00 / CIEDE2000	A colour-difference measure. Always read the number together with the two colours being compared.
TIFF16 / RGB	An image with 16 bits per colour channel; RGB denotes red, green and blue.
TI1, TI2, TI3	ArgyllCMS formats for colour patches, layout/measurement definitions and measurement data, respectively.
JSON / JSONL	Structured text for project data. JSONL stores separate records line by line, such as an event log.
FWA / OBA / BPC	Fluorescent Whitening Agent / Optical Brightening Agent. BPC means Black Point Compensation.
RIP / mesh	Raster Image Processor: software that processes print output. Mesh here means a grid of sample or calculation points.

Standards as references

Project documentation relates quality assessment to ISO, the International Organization for Standardization. References to bvdm mean Bundesverband Druck und Medien, the German printing and media industry association. An internal threshold, such as ΔE00 greater than 5 in a list, must not alone be read as full ISO approval.

Open source and licences

InkProf is distributed under the GNU General Public License version 3 or later. MATLAB and other external components have their own terms. Read LICENSE and THIRD_PARTY_NOTICES in the project for licensing and third-party references.

Basis of this handbook

Both handbooks describe program version 1.1.0 with inner-shell refinement and current workflow updates, document revision 2026-10-13. Version 1.1.0 was released on 12 October 2026; the release tag identifies this source version. Documentation and numerical tests are not new physical validation.

InkProf – source and release information

User documentation

InkProf licence

Third-party code and references

v1.1.1 | MATERIALS AND SHADOWS

More samples in dark colours

Matte paper

Project details > Profiling offers Extra shadow detail for Matte. It is used when Paper finish is Matte. Standard distribution disables this adaptation.

Editable starting values

Dark patch emphasis = 2, Shadow grid emphasis = 1.3 and Extra iteration patches = 48. The first target redistributes the chosen patch budget. The next iteration target adds up to 48 dark training patches without removing colours from the image or C3.

Selection follows the profile black point

The current ICC profile estimates Lab. Selection covers the darkest quarter between model black and white, so it also works when matte paper has a relatively light black. RGB16 duplicates are avoided. Actual count, colours, limits and profile hash are saved in shadow-patches.json.

Measure and compare

Denser shadow table resolution gives less emphasis to lighter regions. This is an InkProf strategy using documented Argyll tools, not an official matte-paper recipe. New patches must be printed and measured. Rebuilding a profile alone does not create new measurement data.

Ink and coating belong to the combination

Specify dye/pigment/mixed/unknown and whether the printer applies coating: off, on, automatic or unknown. Describe the product and settings. These details follow the project and measurement certificate; the app does not change the printer.

v1.1.1 | DELIVERY AND LICENSING

Gamut, certificates and source code

Surfaces and measurement points are different views

View gamut shows an ICC-derived surface in CIELAB D50. It complements measurement/model points and the profile comparison's 2D slice, initially at L*=50. HTML supports rotation; PDF shows a fixed figure. The surface is not a new measurement result.

Deliver a complete report

Save through row 14 after current print verification or row 19 after a scoped numerical decision. Keep the whole delivery folder. Main certificate, Appendices A/B and separate measurement appendix are explained on page 63.

Open source

InkProf code and original documentation are licensed under GNU GPL version 3 or later. Copyright 2026 Christer Törnkvist. The program is provided without warranty under LICENSE. MATLAB, Python, ArgyllCMS, fonts and other dependencies have their own terms; see THIRD_PARTY_NOTICES.md.

i1Pro 2 after the last i1Profiler release

X-Rite has ended support for the i1Pro 2: i1Profiler 3.8.6 is the last version that supports the instrument, and technical support ended in January 2025. Recent Windows 11 releases block the old drivers, and on macOS the software is expected to stop working when Rosetta is removed. The instrument measures as well as before. InkProf uses ArgyllCMS, which has its own instrument drivers, and is therefore independent of further i1Profiler development. The workflow is tested with the i1Pro 2 on macOS. InkProf is an alternative to i1Profiler for anyone who wants to keep using existing equipment with traceable measurement, verification and documentation.

Release status

This handbook describes source version 1.1.0, released 12 October 2026. macOS is tested; Windows/Linux and every instrument mode are not qualified. External runtimes are installed separately.

X-Rite: i1Pro 2 end-of-life and compatibility notes

EXAMPLE IMAGE | VIEW GAMUT

Example: the gamut window

An example of InkProf’s ICC-derived colour volume in a separate window.

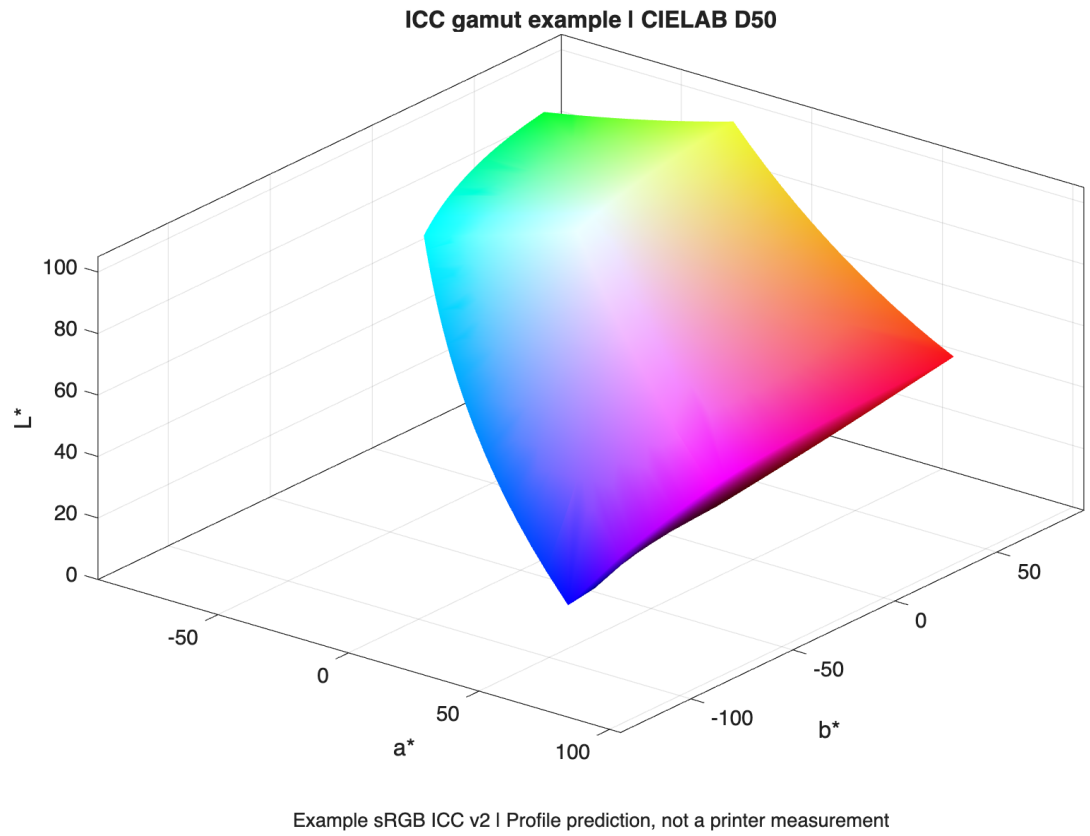


Figure 6. InkProf’s gamut view using an example sRGB profile (ICC v2). sRGB illustrates the window; this is not a measured printer gamut.

Read the axes and surface

L* represents lightness. a* runs from green to red, b* from blue to yellow. The surface is calculated from the profile’s colour transform in CIELAB D50. Rotate the view in the app; the PDF image is static.

Use the view as supporting information

A larger surface is not itself a quality score. Assess measurements, colour errors and real prints. In a printer project, Select area chooses regions for manual reinforcement with additional measured samples. See page 56.

FEATURES AND DETAIL

Two stages of smoothing

Before profile building: Measurement data smoothing

B2 offers Argyll only (raw measurements), without separate pre-smoothing, and Argyll colprof pre-smoothing (experimental), with an intermediate model. Original spectra, XYZ and raw readings are retained. Distinguish this B2 choice from Row 15: Argyll only, which selects additional patches.

Argyll: Assumed noise (%)

The first colprof model uses an assumed average deviation through -r, normally 0.5%. This is not an exact measured Delta E00. The model is evaluated at the original RGB values.

At ICC building: Smoothing at ICC build

This controls final colprof -r. Empty uses 0.5%. The value is independent of the first-stage noise assumption. Percentages display one decimal without rounding stored precision.

FWA/OBA and B1

B2 recipes support native M0 FWA before Argyll pre-smoothing and final fitting. Compensation is applied once, not again to derived XYZ. Changing FWA preserves B1 but requires new B2 and later results. See the M0 white-reference section.

FEATURES AND DETAIL

Test RGB gradients

B2 selection and Manual B3

With pre-regularization selected, tick Open RGB gradient test after profile build. The window opens after successful Manual B3. Save recipe in B2 does not start the check. Automatic B3 creates its own recipes and does not use this B2 choice.

Run RGB check

Nine RGB paths with 1,025 samples each are evaluated through the final ICC. Select a path and relative or perceptual intent. Curves show RGB, Lab, adjacent Delta E00 steps and Lab second differences. JSON and Markdown are saved in the job's checks folder.

Check blue-sky inverse

The blue-sky test uses B2A and returns through A2B. It complements the forward check and changes neither recipe nor ICC. Reopen a closed window with inkprof.checkRGBGradients(jobFolder).

Interpretation

Large local steps, reversals and second differences identify places to investigate. Real printer curvature can also produce these values. No universal pass threshold is used. Printed gradients and documented viewing conditions are needed for physical assessment.

FEATURES AND DETAIL

Remeasure a whole printed row

New measurement or earlier revision

Measure starts a new measurement. Saved revisions selects an earlier saved revision. Import file reads an external file. For isolated problems, use Remeasure patch or Remeasure row in the overview. A new save preserves previous history.

Select page and row

Remeasure row uses the printed page and row number. B24 means position B, row 24; patch ID is separate. Original scan mode, tolerance and instrument conditions are retained. An isolated session measures only the selected row.

Measure and review

Click Measure selected row, calibrate and follow FORWARD/REVERSE. Save supplies candidate values. Change Delta E00 compares previous and new measurements, not error against desired colour. Accept row replacement creates a revision replacing only this row. Discard / Close retains previous values.

Preserved traceability

Original data, new raw readings and the decision are saved. Old paired readings are labelled historical after row replacement. Select and review the correct workflow revision before locking B1 again; Accept in the remeasurement window does not necessarily select profiling input.

FEATURES AND DETAIL

C2: profile, TIFF and print chain

Conversion and tag

Lab-based targets use B2A, absolute colorimetry and BPC off. Layout and RGB16 quantization follow. The exact printer ICC is now embedded as a tag without another pixel conversion. Earlier exports do not change when code is updated.

Keep already converted values

When opening in Photoshop, discard/ignore the embedded profile without converting RGB values; do not convert to a working space. Print at 100% without further colour conversion. Assigned sRGB or Adobe RGB followed by conversion can greatly change older untagged targets. Read PRINTING.txt and document the chain.

Contents and difficult colours

The profiling-workflow default is 80 colours, 24 neutrals, 12 challenges and 12 repeats. Counts can be changed. Challenges may be outside gamut or have an unresolved inverse. C2 therefore does not contain only previous problem colours.

White margin

An unprinted margin uses pure device RGB white. A white image area converted with absolute intent may receive colour to reproduce the reference white point. This is not the paper's own whiteness. A converted image margin and the target background are therefore different things.

FEATURES AND DETAIL

Read the C3 table and warnings

Identity and Lab

Page and Coordinate identify position. ID is identity and Role is patch type. Desired Lab is desired colour; Measured Lab is the print. L* is lightness, negative/positive a* green/red and negative/positive b* blue/yellow.

Two error columns

Measured vs desired (dE00) compares measurement with desired colour. Model vs measurement (dE00) compares ICC prediction with measurement. Similar numbers do not mean similar colours: if both are large, measurement differs from both. Near zero in the second column means close model agreement.

Sorting and groups

Largest errors appear first. Main statistics include unique patches, including challenges, but not extra repeats. Filter groups before drawing conclusions. Small differences between repeats or forward/reverse show repeatability, not accuracy.

Suspected double colour management

C3 simulates extra conversion from sRGB and Adobe RGB to the printer profile and warns when a simulation explains measurements much better. This is a hypothesis; other print faults may look similar. Check assignment, conversion and the driver before retraining the model.

FEATURES AND DETAIL

Reference colours and an existing ICC

Fixed reference set

C2 can use a Lab table, Argyll CIE/CGATS with Lab or XYZ, or TI1 with device RGB. Use only material you are entitled to use. Names and patch codes are preserved for C3. A reference name does not imply protected vendor layout or data are included.

Lab or RGB

Lab references are converted through the ICC and test inverse, print and measurement together. Device RGB is printed as given without ICC conversion; reference Lab is the A2B prediction. PRINTING.txt and verification.json identify the method.

New project: Verify existing ICC

The separate seven-step workflow imports RGB printer profiles in ICC v2 and v4, creates a target, measures, analyses, reviews, records assessment and saves a certificate. The original colour tables are used and delivered. V4.4 is checked natively through LittleCMS; verification does not use a v2 reconstruction. The 575-patch suggestion covers skin tones, shadows, neutrals and broad colours.

Unknown independence

If original training data are unavailable, independence is unknown. Numerical checks and a good photographic impression do not give automatic approval or ISO certification. Document intended use, the chain and limitations.

Methods in RGB Target

The RGB target designer offers OFPS, incremental far point, quasi-random, body-centred cubic and random through Argyll. An existing ICC can guide patch placement but is not measurement data. Argyll preconditioning with v4 needs a separate, explicitly accepted approximate v2 copy; the original is preserved.

FEATURES AND DETAIL

Diagnose with JSON and the right check

Traceable chain

B1 locks measurement input. The job saves recipe.json, status.json and result/profile.icc. C2 saves an exact ICC copy, desired Lab, RGB16 and placement in verification.json. C3 links measurement JSON and its TI3 revision through hashes. Coordinate and ID are different identifiers.

Built-in MATLAB checks

inkprof.checkProfileFit checks the forward model against training. inkprof.checkProfileGrid and inkprof.checkProfileC1 provide grid, CMM and inverse diagnostics. inkprof.checkVerificationTarget compares the verification print. inkprof.checkRGBGradients opens the gradient window. Read each check's purpose.

Separate scripts are not app buttons

The development measurement-JSON inverse script compared Lab, inverse RGB and return through A2B. It is not a built-in InkProf function. Small Lab-RGB-Lab error shows internal consistency, not a correct physical print. Alternative solutions can give large RGB differences.

Test the hypothesis first

First check suspected wrong ICC or inverse numerically and through hashes, without another print. Check a suspected wrong row against TI2 and by remeasurement. A changed print chain may require a new sample with identical device RGB. Do not rebuild the model to conceal an unresolved chain error.

FEATURES AND DETAIL

Compare accuracy and gradients

Open the comparison

After valid B1, choose Compare accuracy and gradients in the main window. Enter one to six positive final colprof -r values. Compare the currently offered Argyll candidates. Several builds may take minutes; progress is displayed.

Read the results

Argyll mode builds one -r 1.0 candidate. InkProf mode compares -r 0.5, 1.0 and 1.5 at high quality when development controls exist. Selection balances development error against float-gradient smoothness and guards against regressions. The default permits at most 0.05 ΔE00 worsening in weighted RMS error for at least 5% gradient improvement. In InkProf mode without development controls it builds one engine-default candidate. These numerical rules are not print acceptance criteria.

Automatic B3 selection

With development controls, Automatic tests final -r values 0.5, 1 and 1.5 at high table quality. Selection balances accuracy and float-gradient smoothness. Guards check patch and gray regressions, curvature, reversals, coverage and colour span. Defaults permit 0.05 RMS accuracy tradeoff for at least 5% gradient improvement. These are numerical selection rules, not print acceptance criteria. Without development controls, one candidate is built with no claim of superiority.

Choose deliberately

Comparison preserves B1 and leaves the current workflow ICC selected. To use a comparison setting, enter it in B2 and build Manual B3. New recipes support FWA before pre-smoothing with the measured-white requirements described in the M0 section.

FEATURES AND DETAIL

Photographic gradient conversion

Open the diagnostic

Choose Photographic gradients in the main window after a successful profile build, or in the RGB gradient window. The check uses the exact selected ICC and saves JSON and Markdown in its checks folder.

What is tested

Nine paths with 1,025 samples are converted from sRGB and Adobe RGB (1998) to printer RGB, then through relative A2B to predicted Lab. Relative and perceptual intent, BPC on/off, and float, 16-bit and 8-bit input/output buffers give 216 combinations. LittleCMS performs these conversions; Photoshop and printer drivers may behave differently.The count is 9 paths × 2 source spaces × 2 intents × 2 BPC settings × 3 precisions = 216; input and output use the same precision.

Read the columns

Curvature P95 measures normalized Lab second differences. L* reversals count decreasing lightness steps on eligible interior points; interpret them in relation to the chosen path. Unique RGB codes and unchanged steps describe quantization plateaus. The JSON also records longest plateau and Lab span. Large numbers are not automatically visible defects.

Limits of the test

Hard device clipping is excluded from comparison metrics, but this is not a proven physical-gamut mask. Quantization plateaus are expected, especially at 8 bit. Compare matching paths and precision. Numerical smoothness does not certify absence of banding; verify printed gradients.

FEATURES AND DETAIL

Certificate evidence

Pre-regularization in the certificate

The certificate records the selected job recipe, pre-regularization method, its assumed noise (avgdev) and final colprof -r. Saved results include the model change when available. Raw measurements are preserved. Model change is not print accuracy.

Gradient evidence and regeneration

A saved photographic-gradient report is included only if its SHA-256 matches the certificate ICC. The certificate summarizes CMM, settings, sample count, interior coverage, reversals and curvature, and embeds full per-path evidence in JSON. Missing evidence is stated explicitly. Re-run step 14 for a new certificate; historical and signed certificates are preserved.

FEATURES AND DETAIL

ICC rendering intents

B2 and rendering intents

Newly built profiles support perceptual, relative colorimetric and absolute colorimetric in one ICC file. B2 provides Perceptual compression (%), passed to colprof -s. The default 20% generic compression is a starting setting, not a measured optimum. Automatic B3 carries forward this B2 value.

What the ICC contains

Perceptual uses a gamut-mapped B2A0 table. Relative uses B2A1. Absolute uses the relative table plus the media white point (wtpt); it does not require a separate B2A3 table. The colour-management application selects the rendering intent. The profile header default does not prevent choosing another intent.Saturation is the fourth ICC intent and uses saturation tables when available. This discussion focuses on the three photographic intents; the version choice on page 46 also concerns saturation.

Build validation and certificate

New builds require perceptual and relative tables with distinct data and a valid media-white tag before publication. Saved jobs and certificates record the rendering-intent settings. Structural validation and numerical conversions do not establish print quality.

Update an existing profile

Build a changed profile with a new B2 recipe and Manual B3. Existing profiles and measurements are retained. Check numerical results and verify the new profile in a fresh print before approval. C2 still uses absolute colorimetry without BPC; print RGB unchanged without further conversion.

ArgyllCMS colprof

ICC specification

VERSION CHOICE | ASSESSMENT AND DELIVERY

ICC v2 or v4.4: which fits best?

Choose according to the printing application, rendering intent and checked results. A higher version number does not itself make a profile more colour-accurate.

What v4 adds

ICC v4 clarifies the reference colour space and rendering intents for more predictable interpretation across profiles and engines. It does not replace good measurements or print assessment.

When v2 is a good choice

Keep v2 in a working, checked workflow, particularly for older applications or recipients requiring it. InkProf uses v2 internally for Argyll. Do not switch solely for a higher version number.

When v4.4 is a good choice

Choose v4.4 when the application and recipient support it and the result has been assessed. InkProf preserves colorimetric tables but remaps perceptual and saturation tables. Shadows and transitions may change; improvement is not guaranteed.

Both for comparison

Both gives both versions from the same profiling run. Compare the same photograph, gradients and shadows with the same paper, ink, settings, intent and viewing light. Record BPC and change one part of the chain at a time.

Numerical BPC example

Two profiles were tested through LittleCMS at sRGB greys 0, 8, 32, 128 and 255. Perceptual v4.4 gave identical results with BPC on and off; predicted L* differed by at most 0.79 from v2 with BPC on. This is a numerical example, not a new print measurement or general guarantee.

What verification establishes

C2 uses absolute colorimetry without BPC. A good result does not establish your preferred perceptual rendering. The certificate identifies the checked profile and v4.4 conversion separately. Also assess a print using the production version and intent.

Import and the practical decision

InkProf imports RGB printer profiles in v2 and v4. Existing v4 verification uses the original through LittleCMS. Argyll preconditioning requires an accepted approximate v2 copy. Choose according to your chain, measurements and image assessment.

ICC: reasons for changing to v4

FEATURES AND DETAIL

TIFF: conversion versus embedded ICC

Read the companion text file

PRINTING.txt beside modern TIFF targets, and *-README.txt for the legacy exporter, list ICC conversion and Embedded ICC separately. Tagged files identify the printer profile and its SHA-256. C2 also has package-level PRINTING.txt; read it together with print/PRINTING.txt.

Conversion: what changed the pixels?

Base and refinement targets render source device RGB directly without a printer ICC conversion by InkProf. Ordinary C2 colour-reference targets have already passed through the printer ICC once, absolute colorimetric without BPC. C2 device-RGB reference sets instead print their RGB unchanged; the ICC predicts reference Lab only. Imported source history may be unknown.

Tag: what identifies the file?

Both kinds of C2 TIFF embed the printer ICC as an identifying tag. Embedding alone changes no pixels and is not an instruction to apply the ICC again. A tag does not prove that no other conversion occurred earlier. InkProf records its own processing separately. Base and refinement targets normally have no embedded profile.

Print without another conversion

Print at 100% with colour management disabled in the application, OS print path and driver. Preserve any embedded tag without conversion. Do not assign another profile or convert the TIFF to a working space. A program can use the embedded tag if colour management is enabled. New packages contain the clearer instructions; existing TIFFs and packages are preserved.

FLOW DIAGRAM | COMPLETE PROFILING

Appendix: workflow diagram

Overview of the example on the following pages: from planning to measured final verification and saved delivery.

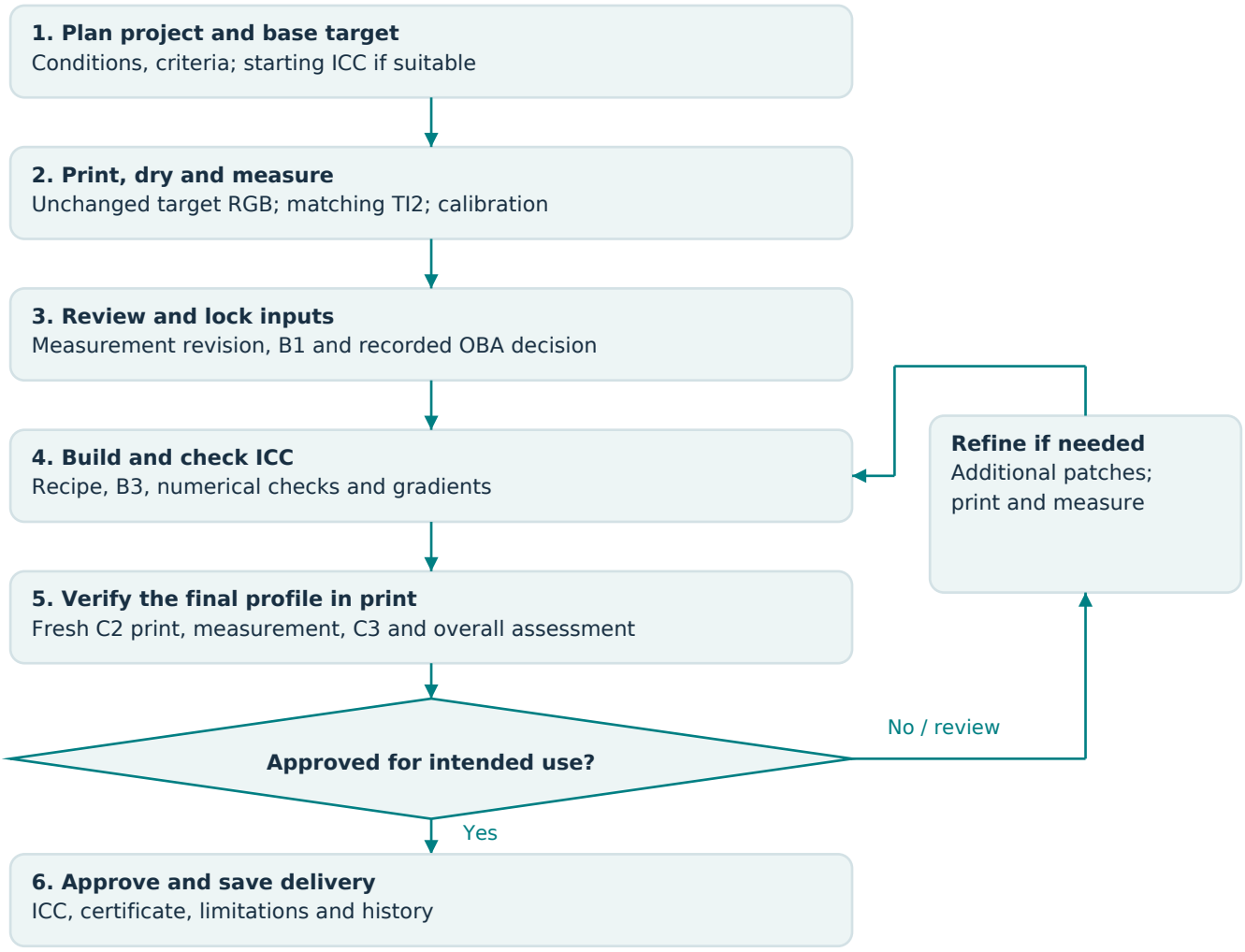


Figure 7. Example workflow. Decisions use current evidence.

Using the feedback loop

When results deviate, first review measurements, profile identity and print settings. Refine only when supported by evidence; otherwise investigate or withhold approval. A new ICC requires fresh final verification. The diagram shows the print-verified route, not the separate numerical delivery route.

COMPLETE PROFILING APPENDIX

Appendix: plan and create the base target

1. Define success before printing

Agree on photographic use or a specific reference condition, viewing light, important colours and gradients, and acceptance criteria. Record printer, ink, paper, driver preset, quality, resolution, backing and instrument. Check nozzles and allow consistent drying time. A custom RGB target does not establish ISO proofing conformity.

2. Use an existing ICC as a starting estimate

Choose a reliable RGB printer ICC for the same printer, ink, paper and settings, or a reasonably similar system. In target generation, select an Argyll method supporting perceptual placement and supply the preconditioning profile. Argyll uses its predictions to distribute device-RGB patches; the base target is not converted through that ICC. A poor or unrelated profile can bias sampling. Without a suitable ICC, use an unconditioned initial target and improve the sampling after the first build.

3. Choose the measurement budget

For careful profiling, plan about 1,500-2,000 patches when time permits. About 800-1,000 is a lower-budget start. A 300-600 pilot checks setup. The design-window initial value of 575 is a smaller starting budget, not a general quality recommendation. The total includes controls and repeats. Inspect the actual count in preview. This is practical guidance, not a universal optimum.

Decision before saving

Preserve patch size and instrument scan geometry. Choose paper dimensions and inspect the actual page preview. Save the TIFF16 together with its matching TI2, layout and PRINTING.txt. Keep the original ICC and its identity with the project. A target-generation preconditioner is distinct from the profile later built from the measurements.

COMPLETE PROFILING APPENDIX

Appendix: print, measure and decide on OBA

4. Print the prepared RGB target

Follow PRINTING.txt. For a missing-profile warning in Photoshop choose Leave as is (don’t color manage). For an embedded profile that must be ignored choose Discard the embedded profile (don’t color manage). Disable further conversion and driver colour adjustment; print at 100%. Do not convert the target to Adobe RGB or apply the printer ICC again. Prepared C2 RGB must also reach the printer unchanged.

5. Measure and review before B1

Calibrate the instrument, use consistent backing and scan direction, and measure all pages. Save each measurement round as a revision. Review identities, white controls, duplicate colours and unusual readings; remeasure suspect patches or a whole row with the correct page and forward/reverse choice. Correct acquisition errors before smoothing. Select and accept the intended revision, then lock B1. Keep different print rounds identifiable.

6. Decide whether FWA/OBA compensation is needed

Do not enable compensation solely because paper looks cool or bluish or a UV lamp makes the paper glow. Consider the paper, verified measurement condition and intended viewing light. For the supported i1Pro2 M0 route, compensation is a D50 simulation using available visible spectra, not a measurement of UV excitation or an M1 result. If the paper has little OBA, or the available evidence does not support compensation, retain the uncompensated route and document the choice.

If compensation is selected

Use native M0 spectra with verified instrument metadata and sufficient spectral coverage. InkProf averages available white-patch spectra and records their count and variation. If no usable white reference exists, B2 offers blank-paper measurement, a saved approved reference, or Cancel. Measure several clean locations on the same paper with the same backing. Averaging can reduce random noise; it cannot recover missing excitation information. Save a new B2 recipe. Keep compensation and viewing assumptions consistent through comparisons and final verification; changing this choice invalidates later results.

COMPLETE PROFILING APPENDIX

Appendix: build, refine and stop

7. Choose smoothing and the B3 route

Start with Argyll only (raw measurements) when repeatability is good. If noisy measurements or gradients justify it, compare Argyll colprof pre-smoothing (experimental) against the raw route. Its Assumed noise (%) and Smoothing at ICC build are separate settings; 0.5% is the initial recipe value, not an established optimum for every instrument. Check colour fit and equivalent photographic gradient paths. Select the RGB gradient test when useful. For an exact saved B2 recipe, choose Manual in B3; Automatic creates its own recipes and does not inherit every B2 smoothing setting. With FWA and pre-smoothing, use the newly saved supported two-pass recipe.

8. Select the refinement method

Use Argyll only as the initial refinement choice: Argyll places additional patches using the current profile. Current InkProf (Jacobian sampling) uses measured local residuals and local response to propose additional RGB samples. Both routes build the resulting ICC with Argyll. Choose Jacobian sampling when a reproducible local discrepancy warrants targeted investigation; do not expect it to make unattainable colours printable. Save the choice with the proposal. Print and measure the added device-RGB target unchanged, then build the next iteration.

9. Compare equivalent evidence

Use row 18 for Previous profile prediction vs current profile prediction at common samples. Separately review fitting error, development/held-out evidence where available, repeatability, gradients and Measured print vs desired colour. A smaller training error alone is insufficient. A small inverse round-trip error only shows consistency of the model and inverse. Large measured discrepancies require checks of profile identity, RGB values, print management, patch mapping and measurement conditions before attributing them to gamut.The C3 table uses Measured vs desired (dE00) and Model vs measurement (dE00). The certificate uses Measured print vs desired colour. Profile comparison uses Previous profile prediction vs current profile prediction; this compares calculations, not a new print.

10. Decide whether another iteration is worthwhile

Continue when a specific measured uncertainty can be reduced. Stop when comparable errors and gradients show no useful improvement beyond repeatability, or a further round produces an unacceptable accuracy/smoothness trade-off. One or two comparable rounds without meaningful gain are a practical review trigger, not an automatic InkProf acceptance rule. Record the evidence, selected profile and reason for stopping. Numerical convergence does not prove physical colour accuracy or compliance.

COMPLETE PROFILING APPENDIX

Appendix: final verification and certificate

11. Verify the exact profile to be delivered

After choosing the final iteration, run row 8 (Fit, Grid and C1) and row 9 (C2: save verification TIFF16) for that current profile. Make a new control print with the documented settings and no further colour conversion. Measure it in row 10, retain the revision, and run row 11 (C3) and row 12 (feedback). Do this even if development or training measurements already look good. A measurement used to train a later build cannot independently verify that later ICC.

12. Assess the whole result

Review count, mean, median, P95 and maximum for all assessed unique colours, not only the worst patches. Identify challenge colours, controls and repeatability. Compare desired-colour error with current-profile prediction error where available. Inspect important photographs and gradients. Model-reachable, outside-model/inversion-unresolved and unclassified references describe model evidence; they are not proof of a physical gamut limit. Confirm white, black, neutral balance and the intended viewing conditions.

13. Approve or decline the intended use

In row 13, enter review notes, scope, print settings, limitations and the decision. Approval is justified only by the relevant measured evidence and agreed criteria. If a required standard limit is not met, report it; printer limitations do not waive that requirement. Photographic use may be approved with explicit limitations when agreed, while conformity for a different use is declined. If the remaining cause is unresolved, say so. Row 14 exports the selected ICC and English measurement certificate.

14. Preserve the evidence and status

Check that the certificate identifies this ICC, iteration, current verification revision, regularisation, FWA reference and reviewer decision. Retain the target, TI2/layout, PRINTING.txt, raw measurements, recipe, ICC and complete reports. Row 19 is numerical-only delivery and cannot replace fresh measured verification. If the ICC or print setup changes after approval, repeat final verification. The defensible conclusion is measured and approved for the stated use, with disclosed limits—not automatic ISO approval.

COMPLETE PROFILING APPENDIX | SOURCES

Appendix: evidence behind the recommendations

Argyll: sampling and measurement quality

The targen documentation supports using a prior or similar ICC to guide perceptual patch placement. printtarg explains instrument-dependent patch geometry and randomised locations, which help identification and distribute spatial printing effects. Preserve the generated layout; do not rearrange patches manually. There is no universally optimal patch count. The suggested budgets in this appendix are InkProf working recommendations to be evaluated against repeatability and independent results.

Argyll: smoothing is an uncertainty assumption

colprof defines -r as the assumed average deviation of device and instrument readings, with a default of 0.5%. Raising it gives less weight to individual readings and more to collective behaviour. A percentage is not ΔE00. Estimate the need from repeat readings and compare accuracy and gradients; do not increase smoothing simply to conceal a systematic print error.

X-Rite: distinguish M0 compensation from M1

X-Rite describes M1 as the preferred condition for OBA-enhanced substrates and explains that M0 does not fully define UV content. The i1Pro2 hardware supports M1, but the InkProf path described here uses verified M0 data. Recording spectral bands near or below 400 nm does not measure the illumination spectrum or establish calibrated UV excitation. Averaging white readings improves repeatability, not the physical definition of M0. Use actual M1 measurements in a compatible workflow when required by the reference or conformity claim.

Fogra: verification needs the relevant reference and control

Fogra links contract-proof verification to its defined MediaWedge, printing condition and applicable tolerances. An InkProf custom RGB verification and reviewer approval are useful evidence for a stated use, but do not substitute for that complete procedure. The independent final control print in this appendix is an InkProf quality-control recommendation.

What external sources do not establish

These sources do not demonstrate that InkProf Jacobian sampling outperforms Argyll only, nor that the recommended patch budgets or stopping triggers are universally optimal. Those decisions require comparable local results. Sources reviewed on 8 October 2026; the handbook describes current InkProf behaviour separately from external guidance.

Argyll: target placement and preconditioning

Argyll: patch layout and randomisation

Argyll: smoothing and profile building

X-Rite: measurement conditions and optical brighteners

Fogra: MediaWedge and contract-proof verification

COMPLETE PROFILING APPENDIX | COMPARISON

Appendix: comparing profiling methods

A photographic starting point

Anders Torger recommends preconditioning, optimised placement and extra greys. His Colormunki example favours about 840 patches as a practical balance; this is experience with that instrument and print system, not a universal limit. He uses colprof -r1.0 to trade a little fit accuracy for smoothness and evaluates diagnostic gradients and a familiar photograph. His reported profile-matching errors compare predictions with measured device patches, including training-data checks; they are not interchangeable with InkProf C3 errors against desired colours on a separate profile-converted print. His tutorial is primarily a base-profiling workflow, not evidence that InkProf refinement replicates a particular iterative algorithm.

A more extensive measurement workflow

X-Rite's recommended RGB workflow proposes 2,033 or 1,586 patches, wider patches for better sampling, about an hour of drying, instrument calibration and saved measurement data. These recommendations concern its own software and layouts: do not blindly increase InkProf patch width or assume one hour guarantees stability for every paper. i1Profiler also offers targeted optimisation and optical-brightener compensation; the internal engine is not Argyll, and its OBC must not be described as identical to InkProf's M0 method.

Reconciled recommendation for InkProf

For careful profiling, plan about 1,500-2,000 patches when time permits. About 800-1,000 is a lower-budget start. A 300-600 pilot checks setup. The design-window initial value of 575 is a smaller starting budget, not a general quality recommendation. The total includes controls and repeats. Inspect the actual count in preview. This is practical guidance, not a universal optimum. Use a suitable starting ICC, neutral/white controls and reliable geometry. Select later refinement from measured evidence.

The final decision remains measured

Neither an attractive photograph nor low model-fit error alone establishes measured accuracy for the intended reference. Keep the familiar photographic check and gradient review, then make the independent final C2 print and C3 measurement before reviewer approval and certificate delivery. This combines practical photographic assessment with traceable measurement.

Torger: Inkjet printer profiling with Argyll and Colormunki

X-Rite: Recommended RGB Printer Profiling With i1Profiler

X-Rite: i1Profiler features and targeted optimisation

COMPLETE PROFILING APPENDIX | VISUAL ASSESSMENT

Test images

Photographic test images

Keith Cooper (Northlight Images) provides printer test images and assessment notes, including the Datacolor image made available courtesy of Datacolor. Use it for overall photographic assessment of skin tones, neutrals, shadows and saturated colours. In InkProf comparisons, inspect the sky gradient for banding or uneven transitions when assessing smoothing; it is a visual diagnostic, not proof of a particular cause. Compare the same source, ICC path, intent, BPC, print preset and viewing light. The original Adobe RGB photograph needs one intended printer-profile conversion; an already converted device-RGB version must receive no second conversion. Do not apply fitting-target opening instructions to the original photograph. JPEG quantisation and clipping can also produce visible steps; use synthetic high-precision gradients alongside it. Download from Keith Cooper’s page; the image is not distributed in the InkProf repository.

Keith Cooper — Printer Test Images, Northlight Images

REFINEMENT | STEPS 15-17

Manual points and an inner shell

Refinement tools is the shared entry point. Choose Manual gamut points / inner shell. The current profile and C1 checks must be complete.

1. Choose a method

Whole inner gamut shell defaults to the entire surface. Selected inner shell areas restricts it to Lab regions around clicked vertices. Selected RGB neighbourhoods adds device-RGB neighbours. Select area enables picking; Rotate view restores rotation.

2. Set thickness and budget

Shell thickness is ΔE76: shortest geometric Lab distance inward from the surface. Default 3, maximum 50. Near-surface lines may guide thickness; there is no automatic detection. Minimum spacing uses RGB percentage points; Maximum new patches is an upper limit.

3. Review the visual check

Review patches / create TIFF16 shows proposed points against the 3D gamut and their depth against shell thickness. Uncheck unwanted patches. These are ICC predictions, not measured print errors. Selection prioritises gaps in modelled colour coverage.

4. Create and copy TIFF16

Review paper dimensions and choose Create TIFF16 target. Status shows phase and elapsed time, including copying and checksum checks. Select another destination or cancel the copy chooser to retain only the project package. Close exits the gamut view.

5. Register the exact print in step 15

After closing, step 15 returns to registration. For already printed sheets: From saved gamut selection, select the correct proposal.json and Use existing print package. Record your reason. The same TI2 and patch order are preserved; pending C2 patches are not added. Create new print package requires a new print.

6. Measure and rebuild

Step 15 must be Complete before step 16 unlocks. Print at 100% without colour conversion and allow drying. Measure the matching TI2 in step 16 and build the next iteration in step 17. Assess it with a fresh independent verification print.

The shell follows the estimated ICC gamut. Extra measurements may improve the model but cannot expand physical printer gamut. ΔE76 is geometric thickness, not ΔE00 or measured print error.

BEFORE STRIP MEASUREMENT | RGB PRINT CHECK

Catch incorrect print settings early

InkProf checks the RGB squares on new pages before the long strip measurement starts. A large difference from an approved print can stop an incorrectly configured print after just a few spot readings.

1. Check settings even with Photoshop presets

A saved Adobe Photoshop preset is not proof that the current print uses the correct settings. Check colour management in Photoshop and the printer driver, media type, resolution and other project conditions before every print. Base targets and already prepared C2/refinement targets must not receive an extra ICC conversion. Follow the target printing instructions.

2. Measure the three controls before the strips

Every TIFF16 page has R, G and B at the upper right. Nominal 8 × 8 mm squares expand by one pixel on every side, with clearance for the caption. Older targets place them below. Calibrate i1 Pro 2 as instructed; measure stationary in M0 on the same backing. Measure each square ONCE: three readings per page. Use the program button, not the instrument button; do not swipe. Confirm the next sheet at Change printed page. All pages are checked.

3. Approve a known-correct print as the reference

The first check needs a print whose paper and colour settings you have independently checked. After the spot readings, select Approve as print reference. InkProf saves the RGB values as the project reference. Without an approved reference, pages are compared with page 1; this cannot detect a shared error on all pages. Changed recorded printer, paper or print conditions require a new reference.

4. Excessive deviation: Remeasure

Above the review threshold, normally 5 ΔE00, the page, colour and deviation are flagged. Select the row and Remeasure page / colour to read that square once again. The previous value remains in history. Restart all RGB checks repeats the sequence. If the deviation persists, check paper and print settings and reprint as needed. Override… requires a written reason; missing readings cannot be bypassed.

Separate from profiling; older pages still work

Control squares do not affect profile fitting. Older targets without declared RGB controls are measured as before and are not rejected for missing squares. Recorded Paper finish, such as Matte, is retained with the check. Compare matte papers with a correct reference for the same paper; a smaller gamut does not itself mean a larger difference from that reference. Paper type and reasoned exceptions may inform future thresholds, but the threshold does not change automatically.

STEPS 9-14 | VERIFICATION AND APPROVAL

From verification target to an approved certificate

A good measurement result is not a saved approval. Verify the current ICC, record your assessment and then export the certificate.

1. Row 9: choose a target for the current profile

Run row 9. Choose Create new target, then Default or Other file for ColorChecker (a Lab file, not an image), or Generated balanced set for broader control coverage. 12 extra repeats means 12 extra patches. For an already printed ColorChecker, choose Select existing target; see page 59.

2. Print the saved TIFF16 target

Print at 100% size with the recorded paper and settings. Lab references have already been converted through the ICC once. Do not apply another colour conversion in Photoshop or the driver. Check active settings even with presets. Allow the print to dry. Page 57 describes the RGB check before strip measurement.

3. Row 10: measure the correct print and save

Run row 10 and choose Measure for a new measurement. Check the target name and patch count in Confirm measurement target before choosing Measure this target. Save and finish saves the measurement. Saved revisions selects an already saved measurement of the same target. Previous files remain when the C2 selection changes.

4. Rows 11 and 12: compare and review

Run C3. Compare model and print, then Review feedback. Assess measured errors, grays, repeats and out-of-gamut colours. Approve only for the use and requirements supported by the results. Evidence from an older target or profile does not replace verification of the current ICC.

5. Row 13: save your explicit approval

Run Approve for intended use. Enter intended use, quality requirements and accepted limitations in Your assessment. Tick the confirmation that you reviewed the results and checked print settings against the project. Select Approve for stated use. Example: Approved for photographic prints on the stated paper after ColorChecker verification. Measured colour errors and stated limitations are accepted for this use. Adapt the text to your actual assessment.

6. Row 14: export the certificate

Run Save ICC and measurement certificate. The certificate includes current results and your approval tied to the profile. ColorChecker has its own section with patch names, reference and measured colours, and Delta E00. This is not automatic ISO certification. Two targets are not automatically combined; the current C2 verification provides the evidence.

STEPS 9-10 | RESELECT A PRINTED TARGET

Measure your already printed ColorChecker target

Select a saved verification target for the current ICC without creating or printing it again.

1. Select the target in step 9

Select C2. Save verification target as TIFF16 and click Run selected step. Choose Select existing target. The list shows saved targets in this project for the current ICC, with names, patch counts and folder IDs.

2. Match the physical print

Choose the ColorChecker entry matching your sheet. Example: Profile test ColorChecker-SG-D50-Lab | 153 patches. The count depends on the reference, repeats and any paper-white controls; not every ColorChecker target has 153 patches. Also compare the folder ID with your print file if several similar targets exist.

3. The saved print package is checked

InkProf checks ICC identity, TI2, manifest and TIFF16 patch values. The same saved print files are reused; no new layout or colour conversion is created. File checks do not prove that the physical print was made with the correct settings.

4. Begin measurement in step 10

Run C2. Measure / select verification revision and choose Measure. Confirm measurement target shows the target name and patch count. If they match your sheet, choose Measure this target, then Start measurement. Saved revisions is for measurements that are already saved.

5. Save and approve the result

Complete the measurement and choose Save and finish. Then run steps 11, 12, 13 and 14 for comparison, review, approval and certificate export. Changing the C2 target requires current results and a fresh approval; it does not change the ICC profile.

If you printed both ColorChecker and a broader verification target, select the sheet you will measure first. Targets are measured separately. The certificate uses the current, measured and approved C2 target.

DELIVERY | JPG AND SOFT PROOF

Study gradients through the ICC profile

The external delivery folder includes test gradients and a screen preview through the delivered profile.

Open the comparison

Open gradients.html beside the ICC, PDF and HTML certificate. Switch between Original and Soft proof. Select 100 % to inspect without scaling. The ramps include neutral, primary, secondary and blue-sky colours. These are generated test images; this feature does not open a custom JPG.

Which profile is used?

The soft proof uses the exact ICC included in the delivery. The original and preview JPGs contain embedded sRGB. Conversion uses relative colorimetric intent and black point compensation. gradients.json records settings and the profile SHA-256 identity.

Do InkProf profiles work?

A screen preview requires the colour-management engine to use the profile tables for the selected conversion. If it fails, gradients.html states that the preview could not be created.

If the preview is unavailable

A profile may lack support for a colour conversion required by soft proof. The delivery is still saved and gradients.html explains that the preview could not be created. This does not automatically mean the profile is unsuitable for printing.

Save an updated delivery

Save a new external delivery through row 14 or 19. No new measurement is needed solely to add preview assets. Keep the complete folder with evidence.

Soft proof is the profile prediction on screen, not a physical measurement or approval. JPG is 8-bit and may introduce gradient steps itself. Inspect on a calibrated display and retain physical print verification.

TROUBLESHOOTING | PYTHON ON A NEW COMPUTER

If a Python package cannot load

InkProf detects failed Python imports and explains how to repair the selected environment. The profiling algorithm does not need to be changed.

Check the installation first

After installation or an update, run setupInkProf(CheckPython=true) in MATLAB. InkProf checks the Python packages needed for analysis and reports by actually importing them. The check also probes required compiled components, such as those in SciPy. An installed package is not necessarily usable.

What does InkProf do when an import fails?

The affected analysis stops before starting. The message identifies the selected Python environment, the original error and commands to repair it. InkProf never reinstalls packages automatically. The standard measurement bridge can run without analysis packages.

Follow the instructions on the affected computer

Copy the repair commands from the message into Terminal on the computer where the failure occurs. They target the Python used by InkProf. Also run pip check as instructed. Do not copy .venv or local-config from another computer.

Check again and continue

Restart MATLAB and run setupInkProf(CheckPython=true). Once the check passes, retry the operation, for example From image in step 15. If it still fails, retain the complete message and Python path for diagnosis.

Version compatibility

An incorrect or incompatible Python or dependency-package version can prevent analysis from working even when the package is installed. InkProf checks the Python version requirement and tests that required packages and their compiled libraries load. Setup also checks that Argyll tools start and report version information. Use the package versions in the requirements files supplied with your InkProf release. The checks test functionality and basic requirements; they do not replace the package versions specified in those files.

This guidance applies to 1.1.0 with current source updates, document revision 2026-10-13. Older release archives may retain older dependencies. Checks and installation do not change saved projects or measurements.

Details and repair commands

APPENDIX | ICC GRID TRIAL

Try a higher ICC grid

A denser ICC grid is a separate trial from the same saved measurements. It adds no measured information and requires no new chart.

Start from the shared menu

Refinement tools > Try higher ICC grid (same measurements). Choose High or Ultra for B2A or both A2B and B2A, above the original level. Default: B2A: Ultra. Argyll chooses n through quality levels; arbitrary n is not offered.

What changes?

B2A converts to printer RGB. B2A-only retains forward-model and pre-fit quality. Both also changes A2B and pre-fit resolution when pre-smoothing is enabled. Measurements, smoothing value, colourimetry, FWA and mapping follow the original recipe. A separate candidate is saved; the active profile is retained.

Visual numerical verification

Table and bars show actual n and n^3 for source and candidate. Select a test gradient for overlaid printer-RGB curves and predicted L*. Dashed: source; solid: candidate. The same nine sRGB paths use relative colorimetric, BPC off and float LittleCMS. grid-curves.json stores values and ICC hashes.

Assess the benefit

Fit, grid and photographic-gradient checks are saved with the candidate. Training error is not independent validation. A denser grid may describe the model in more detail but cannot replace measurements in sparse regions. Use shell printing and measurement to investigate those areas; physical improvement requires print assessment.

Shell refinement changes the measurement set. Grid trials change computed profile tables. Test separately to understand their effects.

v1.1.1 | WORKFLOW AND CERTIFICATES

Next steps and clear delivery

This page explains suggested next rows and how the delivery documents fit together.

Next step

The app suggests the next step from saved, valid results. After step 15, measure in 16, build in 17 and compare in 18. Go to suggested next step selects the step; it does not run it.

Stopped profile build

When numerical checks stop a build, open step 17 results and inspect the latest C1 and Grid reports. Measurements and the previous active ICC are retained. The candidate is not activated automatically. Review the cause before repeating the build.

Certificate structure

The certificate is in English. Its main PDF contains summary, scope, ColorChecker swatches when available, decision and signature. Appendix B with legal terms is in the same PDF. For print-verified delivery, Appendix A with explanations/references is also in the main PDF; for numerical row 19 that technical part is in measurement-appendix.pdf. The separate measurement appendix contains complete records and technical evidence, with the same certificate ID and ICC hash. Earlier measurements are labelled historical. Keep the complete delivery folder, including evidence assets, for working HTML links.

ROW 1 | RGB GRID AND SETTINGS

First grid: levels and patch budget

Initial levels per axis: what does n³ mean?

For mesh this is the count of equally spaced levels on each of R, G and B. Five levels give 0, 25, 50, 75 and 100 percent: $5^3 = 125$ combinations. Seven give 343 and nine give 729. These are colours to print and measure, not the ICC internal LUT grid.

Maximum total patches: the complete budget

The cap includes fitting patches, Control patches and Extra repeat patches. The initial values of 575 total, 64 controls and 40 repeats leave 471 fitting places. Extra gray levels add neutral RGB points; levels already in the mesh are not counted twice. Argyll methods do not use Initial levels per axis as an n^3 starting grid. Nine initial levels require at least 729 fitting places before extras, so they cannot fit a total budget of 575.See page 49 for larger profiling budgets.

Control patches and Extra repeat patches

Controls are separate from fitting data and provide control/development evidence. Repeats are additional measurements of selected fitting colours and help assess repeatability. Neither setting means the three large R/G/B squares checked before strip measurement.

Preview initial grid and refinement

The mesh preview shows the starting grid and RGB distances. Geometric refinement then fills sparsely covered areas within the budget. Distances are in RGB, not measured ΔE00. Choose a budget appropriate to measurement time and paper layout, and inspect the actual generated count before saving.

ROW 1 | METHOD, PART 1

First-target methods: mesh and Argyll

InkProf mesh refinement

Starts with a regular RGB grid and fills geometric gaps in device RGB. You control levels, budget and placement. No preconditioning ICC is used. Useful for explicit RGB coverage; geometric uniformity does not mean equal Lab colour differences.

Argyll OFPS (default, adaptive)

Optimises point spread; a preconditioning ICC can guide adaptation. A natural Argyll choice for a broad initial target.

Argyll incremental far point

Adds points successively far from existing points. A different spreading procedure from OFPS.

Argyll quasi-random, device space

Uses a low-discrepancy distribution in RGB, seeking space coverage without a regular n^3 grid.

Argyll quasi-random, perceptual space

Uses that distribution with perceptual placement. A relevant preconditioning ICC provides a more useful distance basis.

What does default mean?

OFPS is Argyll’s default, but InkProf’s design window normally starts with mesh. Active matte shadow emphasis selects OFPS and locks the method. No method is universally best for every print system.

Argyll targen: methods and preconditioning

ROW 1 | METHOD, PART 2

First-target methods: remaining choices

Argyll body-centred cubic, device space

Uses a structured cubic pattern in RGB, with points between the grid corners.

Argyll body-centred cubic, perceptual space

Uses corresponding structured distribution with perceptual placement based on colour response.

Argyll random, device space

Uses random placement in RGB. A small target can contain local clusters and gaps.

Argyll random, perceptual space

Uses random placement in perceptual space, rather than directly in RGB.

Pre-conditioning ICC and Optimized points (-G)

Choose a suitable profile for the same or a similar print system when available. It guides placement; it does not convert the base target or replace new measurements. OFPS and perceptual methods use it when supplied. -G requests more optimised OFPS placement and may take longer; it does not automatically improve every method.

How should I choose?

Use mesh to control an explicit RGB starting grid and geometric gaps. Use OFPS for broad Argyll placement, preferably with a relevant profile. Quasi-random offers another spread; BCC adds structure. Random is mainly an alternative for comparison. Assess reliable measured results rather than choosing by the method name.

Argyll targen: methods and preconditioning

ROW 1 | INTERIOR PLACEMENT

Mesh: placement and stopping conditions

Centroids (default)

Chooses tetrahedron centroids between existing RGB points. The candidate is inside its tetrahedron. The simplest starting choice.

Sphere centers in tetrahedra

Allows circumsphere centres when they lie inside their tetrahedron. Centroids are also retained as candidates. This seeks gaps without using sphere centres outside the tetrahedron.

All interior sphere centers

Allows centres outside their own tetrahedron if they lie inside the RGB cube. Produces different candidates and coverage. Interior refers to the RGB cube, not a Lab gamut shell.

Max interior gap (0 = off)

A geometric distance condition in normalised RGB, with channels from 0 to 1. Refinement can stop once the largest candidate distance reaches the limit. Zero disables this condition. It is not a ΔE threshold.

Initial gap ratio (0 = off)

Looks for a jump between sorted initial distances and can use it as a stopping level. A value greater than 1 is needed to enable the comparison. Zero disables it. Both gap settings apply to mesh, not Argyll’s spread algorithms.

Check before printing

Inspect placement, actual count and the reason refinement stopped. Preserve the matching TI2 and patch order of a saved TIFF. Distinguish initial geometric mesh from later error-guided Jacobian refinement and shell refinement.

ROW 15 | LOCAL COLOUR RESPONSE

Jacobian: why it matters

A local description of model sensitivity

The Jacobian is a 3 × 3 matrix describing how the profile’s predicted L*, a* and b* change when R, G or B changes slightly. Columns represent RGB channels; rows represent Lab components. It distinguishes sensitive directions from directions where RGB changes produce little Lab response.

A simple worked example

Suppose the R column is [0.20; 0.60; -0.10]. Increasing R by one percentage point with G/B unchanged estimates +0.20 L*, +0.60 a* and -0.10 b*. Two percentage points give roughly twice that change, only while the local linear approximation holds. These are teaching values, not measured printer data.

How InkProf computes it

Current code probes the profile forward transform near the RGB point, normally at ±0.5 percentage points per channel. At 0 or 100 percent the probe is restricted to the permitted range. The predicted Lab difference is divided by the actual RGB difference.

Reading the measures

Units are Lab units per RGB percentage point. The norm of predicted local Lab change is geometric sensitivity, not ΔE00. Singular values describe strong and weak directions. A high condition number indicates uneven sensitivity; nearSingular records an almost lost direction.

Model and physical measurement

The matrix comes from the ICC model and inherits its limitations. C3 measurements establish whether the model differs from the print. A high condition number is diagnostic, not by itself a stopping rule or proof of a faulty printer.

ROWS 15-18 | ERROR-GUIDED REFINEMENT

Jacobian: use and review

What the algorithm actually does

Current InkProf (Jacobian sampling) uses local measured-versus-predicted Lab discrepancies. Candidates are formed between relevant RGB observations and nearby fitting points. The Jacobian estimates the Lab effect of candidate displacement and boosts the priority score with a bounded sensitivity factor.

Why corroboration matters

The algorithm reviews repeats and seeks support from nearby discrepant observations. An isolated error may be flagged for review rather than producing new points. Candidates are quantised to RGB16; already measured RGB values and insufficient spacing are excluded. The patch budget limits selection.

Using the choice

First review C3 and feedback: target identity, colour management, measurement conditions and repeatability. Choose From verification errors and Current InkProf (Jacobian sampling) for a recurring local model discrepancy. Print the proposal unchanged, measure in 16, build in 17 and compare in 18.

What it does not do

The current algorithm does not invert the matrix to correct a patch directly. It does not independently build an ICC or guarantee error reduction. Condition numbers are diagnostic; selection mainly uses predicted local Lab change. More sampling cannot make colours outside the print system’s capability attainable.

Argyll and shell are different choices

Argyll only lets targen place additional RGB points with the current profile. Jacobian refinement combines local measured errors with model sensitivity. Shell refinement seeks coverage gaps near the gamut surface and may be useful without the same C3 error guidance. All three routes use Argyll colprof.

After the next iteration

Compare errors, development evidence, gradients and the earlier profile on equivalent terms. Better priority scores or lower training error do not establish improved print accuracy. Fresh physical verification is needed to assess the new ICC.

B2 / B3 / REBUILD FROM SAVED SPECTRA

Enable OBA/FWA after measurement

No code change is needed. The setting already exists; the profile must be rebuilt.

1. Check the measurement data

Native spectral measurements are required; stored XYZ alone is insufficient. OBA/FWA refers to optical brighteners. The setting enables compensation for their effect, not a new instrument measurement condition.

2. Change the setting

Open Project details and select Compensate optical brighteners (D50). Under Profiling, set Colour data to Spectra (D50 / 2 degrees). Alternatively use B2 > FWA / OBA > Compensate to D50 (also updates Project details). Save the dialog.

3. Rebuild

Save a new B2 recipe in row 6, build B3 in row 7, then run Fit, Grid and C1 in row 8. Manual uses the saved B2 recipe; Automatic creates its own recipes with the project’s FWA choice.

What remains available

Original spectral readings and the frozen B1 input are retained. Changing only the FWA choice normally does not require measuring the base target again. Earlier recipes, profiles and subsequent assessments become out of date. Earlier physical verification does not approve the new profile.

Paper white and verification

If usable paper-white data are missing, B2 offers blank-paper measurement, a saved approved white reference or Cancel. For physical verification of the new profile, perform a fresh C2 print, measurement, C3, feedback and approval in rows 9-14. Row 19 explicitly records the absence of separate print verification.

Limitation

Compensation simulates D50 from M0 spectra; it does not turn M0 into native M1 measurement or prove improved reproduction. Changed materials or printing conditions are a separate issue from changing compensation alone.

MEASUREMENT / NEW IN v1.1.1

Save and resume a partial measurement

For large targets, save completed physical rows and continue later.

Save before stopping

Select Save completed rows / pause. Wait for Completed rows saved - measurement paused before closing. Open results folder locates the session. A save requested during a scan waits for the instrument’s row prompt and excludes the row in flight from the checkpoint.

Whole rows only

Partial rows are excluded. With forward/reverse averaging, both scans are required for a completed physical row. A row with only one scan is omitted from the resume snapshot and measured again in both directions.

Open the saved session

Open the measurement routine for the same target. Choose Resume saved session… and the folder containing measurement-checkpoint.json. The dialog states the completed-row count, page, printed row and direction at which measurement resumes.

Confirm and continue

Choose Resume measurement and confirm Resume same printed sheets. Use exactly the same physical sheets. If you printed again or replaced a page, choose Cancel and start a new session with fresh RGB checks. The checksum identifies chart data, not physical paper.

Follow instrument instructions

Calibrate when prompted and wait for the displayed row instruction before scanning. Saved scan settings and RGB controls are reused. Restart the first unfinished row from its beginning; for paired measurements scan forward, then reverse.

Finish and review

After all rows, select Save and finish and review the measurement as usual. A checkpoint does not complete the workflow step. An unexpected interruption may lose scans since the last checkpoint. Changed chart data, settings or saved readings are rejected.

Included in v1.1.1: automatically tested; real-instrument resume validation remains necessary.

MEASUREMENT / NEW IN v1.1.1

Remeasurement and RGB controls

Continue after rereading

When navigating back within an active measurement and rereading successfully, InkProf selects the following scan pass. For paired measurements follow the forward/reverse instructions for the same physical row. Separate remeasurement of a previously saved row creates a revision that must be reviewed and accepted.

Reuse only for the same print

A new session’s RGB dialog can offer Reuse checks for same print. Select the earlier session folder with approved RGB checks. Target and control-definition identities are checked. Do not reuse for reprinted or replaced pages, even when chart data are identical.

Check every page

For a new print, measure R, G and B on every page. These controls are separate from profiling patches and excluded from fitting. Approved controls are retained when resuming the same saved partial measurement.

Larger squares in newly generated targets

RGB squares gain 0.5 mm on each edge: 1 mm in nominal width and height. The existing one-pixel raster extension on each edge remains. Measurement rectangles and labels follow the geometry. Existing TIFF files are unchanged; generate a new target to use the larger squares.

If RGB checking is interrupted

A cancelled or failed RGB check no longer permanently disables startup. Correct the problem and start again. This does not automatically approve measurements or exceptions.

Included in v1.1.1. Restart InkProf to load the updated code. Existing print files are unchanged.